UNIT – I

Productivity and Work Study

Syllabus (as given)

Productivity and Work Study: definition of productivity, task of management, productivity of materials, land, building, machine and power, factors affecting productivity, work content, basic work content, excess work content, how a manufacturing job is made up, work content due to excess product and process, ineffective time due to shortcomings on part of the management.

Definition, objective and scope of Work Study: work study and management, work study and work.

1. Productivity – Definition

- Productivity is the **ratio of output to input.** It tells us how well the resources (men, materials, machines, money, land, energy) are converted into goods / services.
- Productivity = **Output / Input** (output = units produced / sales value ; input = resources consumed)
- European Productivity Agency: "Productivity is an **attitude of mind** – a mentality of progress, of constant improvement of that which exists."
- Productivity does NOT mean working harder. It means working **smarter** – same / more output with same / less input.

Productivity vs Production

Production = total quantity of output (absolute). Productivity = output per unit of input (relative).

Production can rise while productivity falls (if input rises faster than output).

Efficiency = actual output / standard output (how well a target is met); Effectiveness = doing the right thing; Productivity = efficiency + effectiveness together.

Types of productivity

Type	Formula	Remarks
Partial (single factor)	Output / one input (labour, material, capital or energy)	Easy to compute, but may mislead as other inputs are ignored
Multi-factor	Output / (labour + capital + energy ...)	Considers some inputs together

Type	Formula	Remarks
Total factor	Total output (Rs.) / Total input (Rs.)	Best overall indicator; all inputs in money terms

Common partial productivity measures

- Labour productivity = **Output / Man-hours** (units per man-hour)
- Material productivity = Output / Material consumed (kg, Rs.)
- Capital productivity = Output / Capital employed (Rs.)
- Machine productivity = Output / Machine-hours
- Energy productivity = Output / kWh consumed

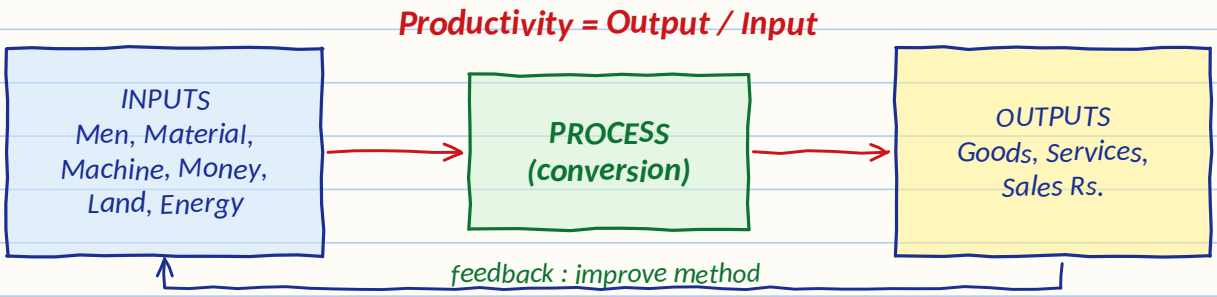


Fig 1.1 Productivity as input–process–output system

Five ways of improving productivity

Case	Output	Input	Result
1	increases ↑	same	Productivity ↑ (most common)
2	same	decreases ↓	Productivity ↑ (cost cutting)
3	increases ↑	decreases ↓	Productivity ↑ ↑ (ideal case, e.g. method study)
4	increases ↑ ↑	increases ↑ (less)	Productivity ↑ (expansion)
5	decreases ↓ (less)	decreases ↓ ↓	Productivity ↑ (retrenchment)

Solved examples

Ex1. A shop makes 4800 units in a month using 1600 man-hours. Labour productivity = 4800 / 1600 = **3 units / man-hour.** After method improvement it makes 5400 units in 1500 man-hours: = 5400 / 1500 = 3.6 units / man-hour. Increase = (3.6 – 3)/3 = **20 %.**

Ex2. Output worth Rs. 6,00,000. Inputs: labour Rs. 1,50,000; materials Rs. 2,50,000; capital Rs. 50,000; energy Rs. 20,000. Total input = Rs. 4,70,000. Total productivity = 6,00,000 / 4,70,000 = **1.277.** Labour productivity = 6,00,000 / 1,50,000 = 4.0.

Ex3. Productivity rises from 2.5 to 2.75 → % increase = (2.75 – 2.5)/2.5 × 100 = **10 %.**

2. Task of Management

- Management's basic task is to **get things done through people** and to make the **best use of the 4 M's** – Men, Materials, Machines, Money (plus Method, Minutes / time).
- Functions: **Planning → Organizing → Staffing → Directing → Controlling** (POSDC).
- To produce goods / services at the **lowest cost,** at the **right quality,** in the **right time** – so that the enterprise survives and grows.
- Management is responsible for:
  - choice of product and its design, specification, standards
  - choice of process, plant, layout, machine and tools
  - purchase and control of materials and stores
  - selection, training and motivation of people, safe conditions
  - planning and controlling production, quality and cost
- Most of the avoidable waste of time and money in a factory (the **ineffective time)** is due to management decisions – hence "the task of management is to reduce the total work content and the ineffective time".

Management cycle (feedback from control to planning)



Fig 1.2 Functions of management

3. Productivity of Materials, Land, Building, Machine & Power

(a) Productivity of materials

- Material productivity = **Output / material input.** Materials are 50–70 % of product cost, so even small saving gives a big gain.
- Ways to improve:
  - proper material selection; cheaper substitute without loss of quality (value analysis / value engineering)
  - reduce scrap, rejects, rework, wastage; increase **yield** (e.g. nesting of blanks, better cutting patterns)
  - standardisation & simplification of materials and parts; bulk purchase, vendor development

- scientific inventory control (EOQ, ABC, JIT), proper storage and handling to avoid damage, obsolescence
- recycling of scrap, use of by-products, proper handling to avoid pilferage
- Yield % = (weight of good product / weight of raw material) × 100

(b) Productivity of land and building

- Land and building are costly and fixed – use them fully. Measure: **Output per sq. m of floor area** (or Rs. per sq. m).
- Improve by: good plant layout; remove unused machines & junk; **vertical storage** / racks; multi-storey buildings; reduce passages and idle space; use of overhead conveyors; proper location of stores.
- Floor-space utilisation = (area occupied by machines, work & aisles in use / total area) × 100.
- Good building: natural light & ventilation, flexible bays, strong floors, easy extension – reduces cost and delays.

(c) Productivity of machines

- Machine productivity = **Output / machine-hours;** Utilisation % = (productive machine time / available time) × 100.
- Improve by: preventive maintenance (reduce breakdown), reduction of set-up time (SMED), correct speeds & feeds, better tooling & fixtures, replacement of obsolete machines, automation, proper machine loading and scheduling, trained operators.
- Overall Equipment Effectiveness: OEE = **Availability × Performance × Quality.**

(d) Productivity of power (energy)

- Energy productivity = **Output / kWh** (or per litre of fuel). Energy cost is a rising share of cost.
- Improve by: power-factor correction (capacitors), energy-efficient motors, switching off idle machines, avoiding peak demand, waste-heat recovery, leak-free compressed air, good insulation and lighting, regular energy audit.

4. Factors Affecting Productivity

- Factors are classified into **internal (controllable)** and **external (uncontrollable)** factors.

Internal factors	External factors
Hard: product, plant & equipment, technology, materials, energy, layout, capital, work methods	Government policies, taxes, tariffs, labour laws

Internal factors	External factors
Soft: people (skill, attitude, motivation), organisation structure, management style, training, wages & incentives, communication, industrial relations	Infrastructure: power, roads, ports, communication
Working conditions: light, noise, temperature, safety, welfare, working hours	Natural resources, climate, geography, social & cultural values, market demand, economy, inflation

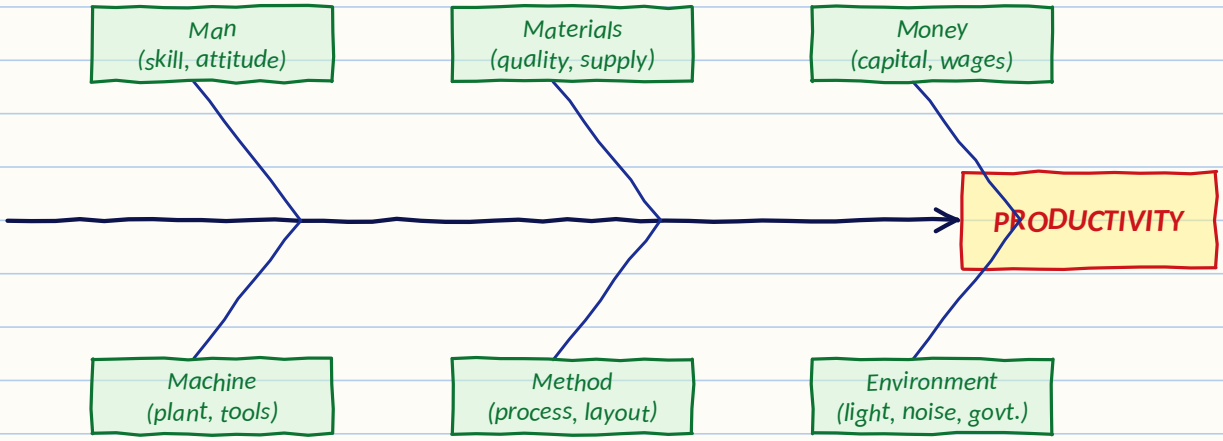


Fig 1.3 Cause-effect (fishbone) diagram of productivity factors

- Other factors: **technology / automation,** quality of work life, **work-study & method improvement,** size of the market, product mix, wage-incentive schemes, TQM, Kaizen, 5S.

5. Work Content

Definition

Work content = the amount of work, measured in man-hours or machine-hours, required to make one unit of a product (to complete the job), when worked at standard performance and with no ineffective time.

Total time to make a product = **Work content + Ineffective time**

Basic work content (BWC)

- The **minimum time theoretically possible** to produce one unit – if the design were perfect, the process perfect and there were no ineffective time of any sort.
- It is the time of the "ideal" product made by the "ideal" process. In reality it is never reached – it is a target to approach (like the speed of light for time!).
- Depends only on the physical / chemical change needed in the material (e.g., the machining time for removing the metal in the most efficient way).

Excess work content (EWC)

- Work content that **adds to the basic work content** because of defects and shortcomings in design, specification or process.
- Two sources:
  - **A.** Work content added by **defects in design / specification of the product**
  - **B.** Work content added by **inefficient methods of manufacture or operation**
- Total work content = Basic work content + A + B

Ineffective time

- **C.** Ineffective time due to **shortcomings of the management**
- **D.** Ineffective time due to **shortcomings of the worker** (within his control)

Equation

Total time of manufacture = Basic work content + A + B + C + D

Total work content = Basic WC + A + B ; Total ineffective time = C + D

6. How a Manufacturing Job is Made Up

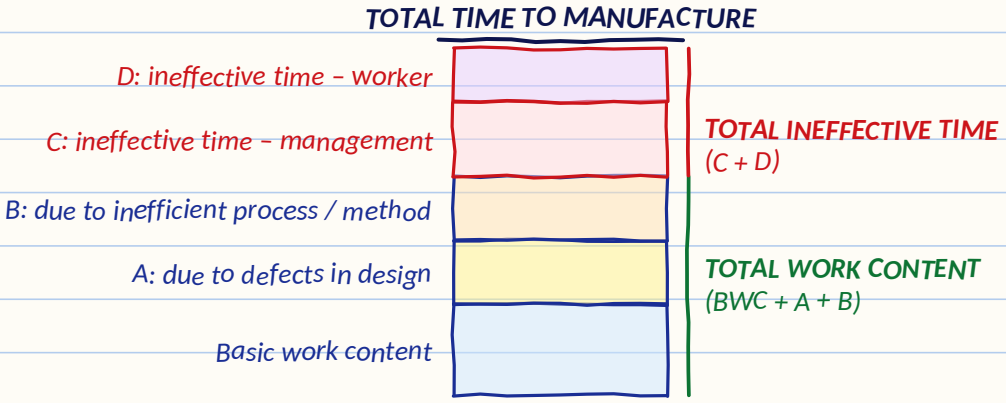


Fig 1.4 Build-up of the total time of a manufacturing job

- The task is to attack the layers from the top: remove D and C first (management action), then B (method study), then A (product redesign) → total time comes down towards the **basic work content.**

Layer	Who can reduce it	Techniques
A (design)	Product designer / R&D / marketing	Value analysis, standardisation, simplification, better specification, design for manufacture
B (process)	Production / process engineer, work-study man	Method study, better tooling, machines, layout, process planning, mechanisation

Layer	Who can reduce it	Techniques
C (management)	Top & middle management	Planning & control, maintenance, stores, training, safe & good conditions, scheduling
D (worker)	Worker with supervisor	Motivation, incentives, discipline, training, joint consultation

7. Work Content due to Excess in Product and Process

(A) Excess work content due to product (design / specification)

- Product design that does not permit the use of the most economical process or method.
- **Lack of standardisation** of parts, materials, sizes – leads to short runs and frequent changeovers.
- Too wide a variety of products / models → small batches, frequent set-ups; products made to individual customer order.
- Unsuitable or too-close tolerances, surface finish, quality standards beyond functional need.
- Too many parts / poor design for manufacture & assembly (DFMA) – more machining, joining, inspection.
- Wrong materials – hard to machine, high waste, heavy; unnecessary stock removal (large machining allowance).
- Frequent design changes; poor liaison between design and production.

(B) Excess work content due to process (method of manufacture)

- **Wrong machines or tools** – e.g. using a lathe where a milling / press tool is faster; worn tools; poor tool design.
- Wrong operating conditions: cutting speed / feed / depth of cut not optimum; wrong coolants.
- Inefficient layout – long transport, back-tracking, waiting between operations; poor material handling.
- Poor sequence and method of work – unnecessary motions, poor workplace layout (no motion economy), poor jigs & fixtures.
- Excess set-up and inspection time, poor process planning, using skilled work for unskilled jobs.
- Poor working conditions – bad light, heat, noise causing fatigue and slow work.

Note: Both A and B can be reduced by method study + value analysis; they are a part of "excess work content".

8. Ineffective Time due to Shortcomings on the Part of Management

- Time during which the **worker is available but nothing useful is produced** because of causes within management control.
- Main causes:
  - **Too wide a variety of products** and frequent changes of design → idle time, rework, set-ups.
  - Failure to standardise materials, parts and methods; bad product design.
  - Poor planning and scheduling → workers wait for work, instructions, drawings, tools, material.
  - Shortage / poor quality of materials (poor purchase & stores control).
  - Machine breakdowns due to poor maintenance; power failure; poor tool supply.
  - Poor working conditions and unsafe practices → accidents, injuries.
  - Lack of training, supervision and motivation; poor industrial relations, strikes.
  - Weak production control, poor layout and material handling → waiting & transport waiting.

Ineffective time due to shortcomings of the worker (D)

- Absenteeism, lateness, idling, slow work (deliberate), careless work causing scrap & rework, ignoring safety rules, taking more rest than permitted.

Note: D is controlled by good supervision, incentive schemes, discipline and training; but the root cause is often a management problem too.

9. Work Study – Definition, Objectives and Scope

Definition (ILO)

Work study is a generic term for those techniques, particularly **method study** and **work measurement,** which are used in the examination of human work in all its contexts, and which lead systematically to the investigation of all the factors which affect the efficiency and economy of the situation being reviewed, in order to effect improvement.

- Work study = **Method study + Work measurement.** Also called "Methods engineering", "Time and Motion study" (older name).
- Aim: to find the **best way of doing a job** and set the **time it should take** – leading to higher productivity with the existing resources (no capital investment).

- Method study → HOW the work is done (reduce work content). Work measurement → HOW LONG it should take (reduce ineffective time).

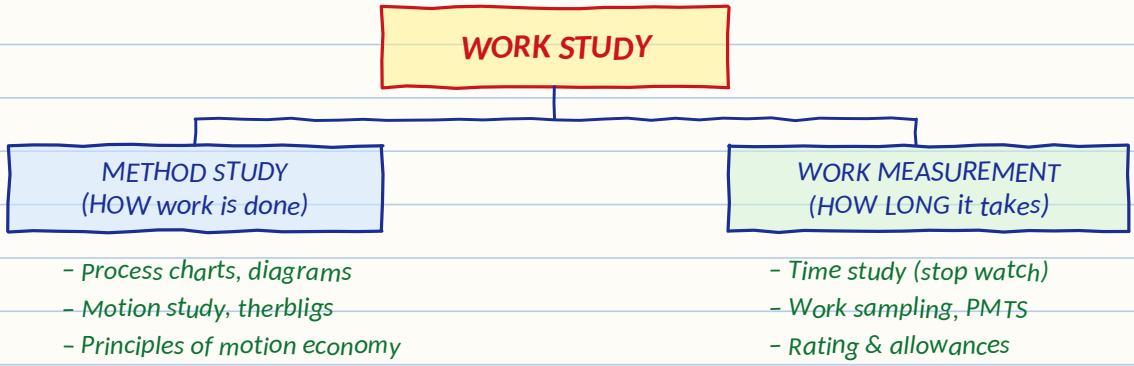


Fig 1.5 Techniques of work study

Objectives of work study

1. To analyse the present method of work and **eliminate unnecessary / wasteful movements** and operations.
2. To develop the **best (standard) method** and standardise it – higher output with less fatigue.
3. To set the **standard time** for the job (for planning, scheduling, costing, wages, incentives).
4. To increase productivity through better use of men, machines, materials and space.
5. To improve plant layout, material handling and working conditions; to reduce fatigue, accidents.
6. To reduce cost, set fair workloads, assess manpower and plan capacity; to give basis for incentive and wage schemes.
7. To train workers in the improved method and to develop co-operation and job satisfaction.

Scope of work study

- Work study is a **universal tool** – applicable wherever human work is done:
  - Manufacturing: machining, assembly, fabrication, inspection, maintenance, stores, packing.
  - Offices & banks (paper work, procedures), hospitals, hotels, retail shops, transport, farms, construction, defence, libraries, software / call centres.
  - Used at all levels: work place, department, whole plant (layout & flow), systems / procedures.
  - Used for new jobs (planning) as well as existing jobs (improvement).

- Why it is attractive: low cost, no or little capital, **systematic,** gives quick results, can be repeated, raises productivity of existing resources.

Basic procedure of work study – "SREDDIM"



- SELECT the job to be studied → RECORD all facts of the present method → EXAMINE critically → DEVELOP the most practical, economic, effective method → DEFINE the new method → INSTALL it as standard practice → MAINTAIN by periodic checks. (Some texts add MEASURE / EVALUATE.)

10. Work Study and Management

- Work study is a **tool of management** – "the servant of management" – used for investigating and improving the use of resources; it is not a replacement for management decisions.
- Management uses work study to:
  - find causes of high work content and ineffective time and remove them;
  - set realistic standards of performance → planning, scheduling, loading, estimating, standard costing, budget;
  - decide manpower, machine requirements and plant layout;
  - control output and cost by comparing actual with standard performance;
  - design wage-incentive plans and fair workloads; support industrial relations.
- Work study gives **facts** to management – it is the "eyes and brain" of the control system. It is a part of industrial engineering and links to quality, maintenance and production control.
- Work study can function effectively only with **top-management support;** the work-study man works as a staff specialist (advises) – the line manager decides and implements.

Work-study man	Line management
Advises, recommends, trains, helps in installation	Takes decisions, implements and maintains the new method
Collects facts, analyses systematically, develops alternatives	Provides resources, authority, follow-up and discipline
Needs tact, patience, integrity, open mind, technical knowledge	Needs to give time, support, consult workers & union

- Qualities of a work-study practitioner: sound technical background, honesty, **tact,** patience, good communication, enthusiasm, ability to inspire confidence, good judgement, physical fitness, sense of humour.

11. Work Study and Work (the Worker)

- Work study studies "human work" – so the worker is the most important partner. Success depends on the worker's **co-operation.**
- Benefits to the worker:
  - easier, less fatiguing, safer way of working; better working conditions;
  - fair standards and wages (incentive earnings), no arbitrary workloads;
  - proper training, job satisfaction, job security through a stronger company;
  - better relations – the standards are based on **measurement and facts,** not on opinion.
- Human problems: fear of job loss / speed-up, fear of change, suspicion of the "stop-watch man", union opposition, loss of skill & status.
- How to win co-operation: explain the aims in advance, consult the workers & union, **involve operators in developing the method,** give credit for ideas, guarantee no retrenchment due to study, behave politely, never criticise the operator, and keep the study open.

Golden rules while studying people

- Inform the supervisor and operator before the study; explain purpose.
- Be honest and impartial in recording; never take "secret" readings.
- Always give credit to the operator for good ideas; never blame him for wastage.
- Share the results with the union & workers; train them in the new method.

Method study vs work measurement – quick comparison

Point	Method study	Work measurement
Aim	Find the best method	Find time for the method
Reduces	Work content (A & B)	Ineffective time (C & D)
Tools	Charts, diagrams, motion study	Stop watch, work sampling, PMTS
Order	Done first	Done after method is standardised

12. Quick Revision & Important Questions

Memory hooks

- Productivity = Output / Input; Types – partial, multi-factor, total.
- Total time = BWC + A (design) + B (process) + C (management) + D (worker).
- Work study = Method study (how) + Work measurement (how long).
- Procedure: SELECT – RECORD – EXAMINE – DEVELOP – DEFINE – INSTALL – MAINTAIN.

Probable exam questions

1. Define productivity. Explain the factors affecting productivity.
2. Distinguish between production and productivity. How can productivity be improved?
3. What is the task of management? Explain how productivity of materials, land and building, machine and power can be raised.
4. Define work content. Explain with a neat sketch how a manufacturing job is made up.
5. Explain basic work content and excess work content; list the causes of excess work content due to product and process.
6. What are the causes of ineffective time due to the shortcomings of management and workers?
7. Define work study. State its objectives and scope. Explain the relation of work study with management and the worker.
8. Explain the basic procedure of work study with a block diagram.
9. Numerical: compute labour productivity, total productivity and % improvement (see Ex1–Ex3).

UNIT – II

Method Study • Motion Economy • Work Measurement • Work Sampling

Syllabus (as given)

Method Study: definition, objective and scope, activity recording and tools – Outline Process Chart, Flow Process Chart, Flow Diagram, String Diagram, Travel Chart, Multiple Activity Chart, Two-Handed Process Chart. Principles of Motion Economy: classification of movements, micromotion study, Therbligs, SIMO chart. Special charts: Cyclegraph and Chronocyclegraph – development, definition and installation of the improved method.

Work Measurement: definition, objectives, techniques. Work sampling – need, confidence levels, sample size determination, conducting a study with problems.

1. Method Study – Definition, Objective, Scope

Definition (ILO)

Method study is the systematic **recording** and **critical examination** of existing and proposed ways of doing work, as a means of developing and applying easier and more effective methods and reducing costs.

Objectives

- Improvement of processes and procedures; improvement of factory, plant and work-place layout.
- Economy of human effort and reduction of unnecessary fatigue.
- Economy in the use of materials, machines and manpower; improved material handling.
- Development of a better physical working environment; **standardisation** of the improved method.
- Simplifying work so that the same output is obtained with less effort and cost, and better safety.

Scope

- Applicable to every activity involving human work – shop floor, offices, stores, maintenance, hospital, banks, construction, etc.

- Studies: the process as a whole (flow of material / men), a single operation, a work-place, or even a single hand motion (micromotion).

Basic procedure (7 steps)

1. **SELECT** the job. Factors: **economic** (high cost, bottleneck, large volume), **technical** (quality problems, rejects, difficult work), **human** (fatigue, accidents, union complaints, boring work).
2. **RECORD** all relevant facts of the present method by charts, diagrams, film. (Direct observation – accurate, complete, concise.)
3. **EXAMINE** the facts critically using the questioning technique.
4. **DEVELOP** the most practical, economic and effective method; consult workers; compare alternatives with costs.
5. **DEFINE** the new method – written standard practice, drawings, charts.
6. **INSTALL** the new method as standard practice – get approval, train, trial run, change over.
7. **MAINTAIN** the standard by periodic review; prevent drifting back to the old method.

Critical examination – the questioning technique

Aspect	Primary question	Secondary question	Aim
Purpose	What is done? Why is it done?	What else could be done? What should be done?	ELIMINATE
Place	Where is it done? Why there?	Where else could it be done? Where should it be done?	COMBINE / REARRANGE
Sequence	When is it done? Why then?	When else could it be done? When should it be done?	REARRANGE
Person	Who does it? Why that person?	Who else could do it? Who should do it?	COMBINE / REARRANGE
Means	How is it done? Why that way?	How else could it be done? How should it be done?	SIMPLIFY

Note: ECRS principle: Eliminate – Combine – Rearrange – Simplify (in this order of priority). Develop the new method using the answers.

2. Activity Recording and Tools

- Recording must be **accurate, complete and concise** – it is the base of the examination. Never trust memory.

Group	Tool	Used to show
Charts – process	Outline process chart, Flow process chart (man / material / equipment type)	Sequence of events in a process, with symbols
Charts – activity (time-scaled)	Multiple activity chart, Two-handed process chart, SIMO chart	Simultaneous activities of men / machines / hands against time
Diagrams (scaled layout)	Flow diagram, String diagram, Travel chart (From–To chart)	Paths of movement on a layout; frequency of movement
Photographic	Cine film, micromotion, memo-motion, time-lapse, video; Cyclegraph, Chronocyclegraph	Fine motions, patterns of movement

Process chart symbols (ASME)

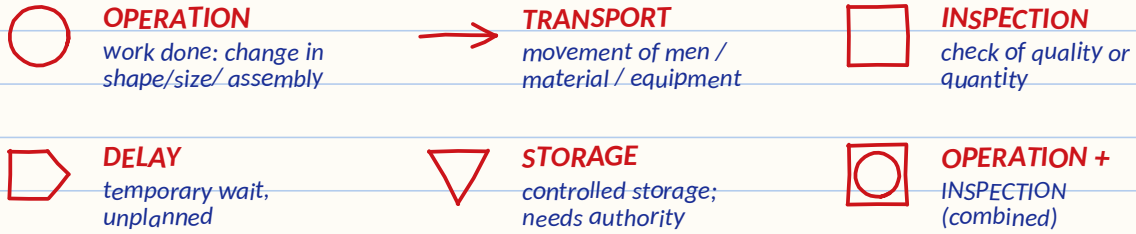


Fig 2.1 Standard (ASME) symbols; size of circle ≈ 10 mm

- Operation = productive step (value-adding). Transport, Delay, Storage = non-productive (try to eliminate). Inspection = non-value-adding but necessary.

(a) Outline Process Chart (OPC)

- Gives an **overall picture** – shows only the **operations (O)** and **inspections (I)** in sequence, and the points where materials enter the process. Also called an "operation process chart".
- Main flow line vertical on the right; material lines enter horizontally; sub-assemblies enter from left. Time can be written on the right.

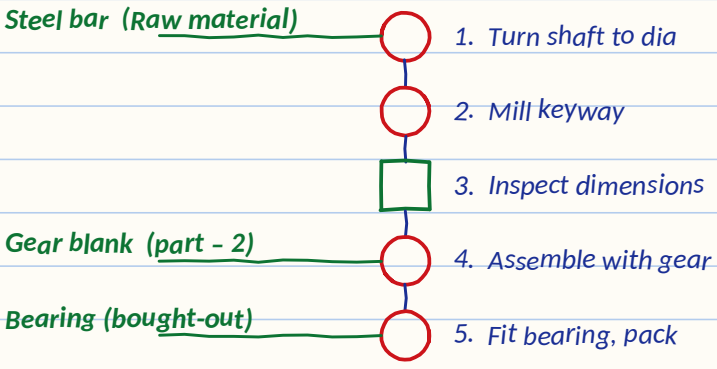


Fig 2.2 Outline process chart for a shaft–gear assembly

(b) Flow Process Chart (FPC)

- More detailed than OPC – shows **all five activities:** operation, transport, inspection, delay, storage; with distance (m) and time (min).
- Three types: **Man type** (what the worker does), **Material type** (what happens to the material), **Equipment type** (how the equipment is used).
- Used for analysing the sequence, to expose transport, delays and storage – then apply ECRS. Compare "Present method" and "Proposed method" with a summary table.

No.	Description of activity	Dist(m)	Time	O T I D S
1	Raw bar in store			
2	Move bar to saw by trolley	25	2.0	
3	Wait for saw		6.0	
4	Cut to length		3.0	
5	Carry to lathe	15	1.5	
6	Turn and drill		9.0	
7	Inspect dia and length		1.0	
8	Move to finishing	20	1.5	
9	Store in bin			

Fig 2.3 Flow process chart – material type (present method)

Summary	Operation	Transport	Inspect	Delay	Storage
No. of events	2	3	1	1	2
Time (min)	12.0	5.0	1.0	6.0	–
Distance (m)	–	60	–	–	–

(c) Flow Diagram

- A **scale layout** of the floor / building on which the path of movement (men / material) is drawn, numbered to match the flow process chart. Shows back-tracking, crossing, long distances, congestion.
- Direction shown by arrows; symbols O T I D S marked at the points of occurrence; different colours for different materials.

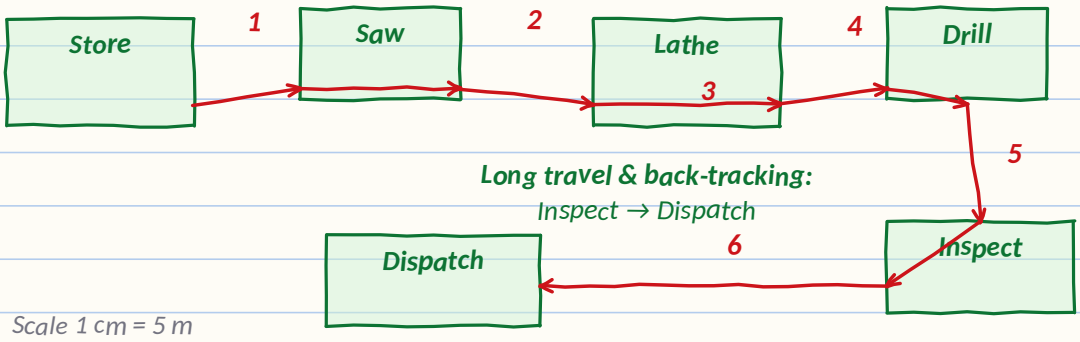


Fig 2.4 Flow diagram of present method (numbers = process chart steps)

(d) String Diagram

- A scale layout with **pins at each work-station;** a continuous **thread** is wound round the pins in the order of the movements. Length of thread used × scale = total distance travelled.
- Used for: movement of a worker / material over a shop floor, repetitive movements, crowded areas – to design a better layout (shorter strings), compare layouts.
- Distance = (string length used) × (scale factor). Heavier string concentration = **more frequent movement** → bring those stations closer.

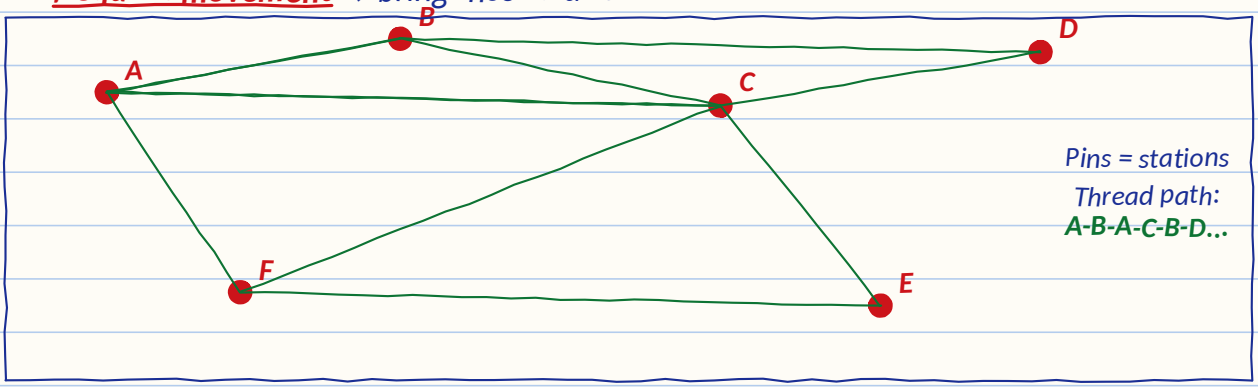


Fig 2.5 String diagram of a work-area

Flow diagram	String diagram
Shows direction & path of flow, single trip	Shows frequency and total travel (repeated movement)
Drawn with lines & arrows; easy and quick	Needs board, pins and thread; more time, more accurate for distance
Good for process flow of material	Good for worker movement, layout of tool crib, stores, offices

(e) Travel Chart (From–To chart)

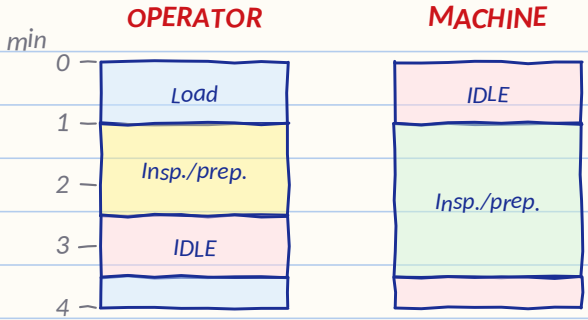
- A tabular record of the **number of movements** (loads or trips) between work-stations / departments in a given period. Used for layout planning – place stations with high travel close to each other.
- Rows = FROM, columns = TO. Frequency × distance gives the material-handling load; the sum of (trips × distance) is to be minimised.

From \ To	Store	Saw	Lathe	Drill	Inspect
Store	–	40	10	–	–
Saw	–	–	35	15	–
Lathe	–	–	–	30	20
Drill	–	–	–	–	45
Inspect	5	–	–	–	–

Note: Highest flows: Drill→Inspect (45), Store→Saw (40), Saw→Lathe (35) – place these pairs adjacent.

(f) Multiple Activity Chart (Man–Machine chart)

- Records the activities of **several subjects (men / machines)** on a common time scale – shows the idle time of each and how activities are inter-related. Used for man–machine and gang (team) work.
- Used to: find idle (waiting) time, improve utilisation, decide the **number of machines one operator can attend,** balance team work.



Cycle time = 4.0 min
Operator utilisation = 3.0/4.0 = 75 %
Machine utilisation = 2.5/4.0 = 62.5 %
Output = 60/4 = 15 parts / hour

Fig 2.6 Multiple activity chart (one man – one machine)

Number of machines per operator (synchronous servicing)

n = (l + m) / (l + w) l = operator loading / unloading time per machine, m = machine running time, w = walking time between machines.

Example: l = 1.5, m = 2.5, w = 0.5 → n = 4.0 / 2.0 = 2. Choose n = 2 (or compute cost per piece for n = 2 and 3 and pick the lower).

(g) Two-Handed Process Chart (Left-hand / Right-hand chart)

- Records the **movements of both hands** of a worker in relation to one another on a time scale – used for repetitive short-cycle bench / assembly work to expose idle hand, holding, unbalanced work.
- Symbols: O (operation – grasp, position, use, release), T (transport – reach, move), D (delay – idle/waiting), (hold). Hold is shown with the inverted triangle.

Left hand	Sym	Sym	Right hand
Reach for bolt	T	T	Reach for nut
Grasp bolt	O	O	Grasp nut
Move bolt to centre	T	T	Move nut to centre
Hold bolt	S (hold)	O	Screw nut on bolt
Hold bolt	S (hold)	O	Tighten with spanner
Place assembly in bin	T	D	Wait / idle

Note: Improvement: use a fixture to HOLD the bolt (frees LH), use drop-delivery bins; both hands then work together.

3. Principles of Motion Economy

- Introduction: Frank & Lillian **Gilbreth** developed motion study; Barnes later formulated the **principles of motion economy** – rules to reduce fatigue and increase output by eliminating useless motions. They are a check-list while developing the new method.

Classification of movements (Barnes / Gilbreth)

Class	Body members that move	Example
1	Fingers only	Pressing a push button, typing
2	Fingers + wrist	Turning a small knob, using a screwdriver
3	Fingers + wrist + forearm (elbow pivot)	Reaching for a tool on the bench
4	Fingers + wrist + forearm + upper arm (shoulder pivot)	Reaching across the bench
5	Class 4 + trunk (body movement)	Bending / twisting to reach a bin on the floor

Note: Rule: do the work with the lowest possible class of movement (class 1 or 2) – it needs least time and fatigue.

- Other classifications: **Fundamental hand motions** (reach, move, grasp, position, release, etc. = therbligs); motions can also be: ballistic vs controlled, symmetrical, continuous curved vs straight-line with sharp turns.

Barnes' principles – three groups

(I) Use of the human body

- Both hands should begin and complete their motions at the same time; and should not be idle at the same time except during rest.
- Motions of the arms should be **symmetrical and in opposite directions** and made simultaneously.
- Hand motions should be confined to the **lowest classification** that can do the work satisfactorily.
- Momentum should be employed to assist the worker; if it must be overcome by muscular effort, reduce it to a minimum.
- Smooth continuous curved motions are preferable to straight-line motions involving sudden sharp changes of direction.
- Ballistic (free swing) movements are faster, easier and more accurate than restricted (controlled) movements.
- Rhythm is essential to smooth, automatic performance; work should be arranged to permit an easy natural rhythm.

(II) Arrangement of the work-place

- There should be a **definite and fixed place** for all tools and materials.
- Tools, materials and controls should be **pre-positioned** and located close in front of the worker – within the normal working area.
- Gravity-feed bins / containers and drop-delivery chutes should be used to deliver material close to the point of use.
- Materials and tools should be arranged to permit the best sequence of motions.
- Provide adequate illumination; chair and work height should permit alternate sitting & standing; chair of proper design for good posture.
- Eye-fixation should be reduced: visual displays and parts in the normal vision field.

(III) Design of tools and equipment

- Use a jig, fixture or foot-operated device to **relieve the hands of holding** work – hands then do useful work.
- Two or more tools should be combined wherever possible (e.g. combination pliers).
- Tools and materials should be pre-positioned whenever possible (ready for the next motion).
- Where each finger performs a specific movement (typing), the load should be distributed according to their capability.
- Handles (e.g. crank, screw-driver) should be designed to give as much surface contact as possible with the hand; levers, cross-bars and hand wheels should be positioned for maximum mechanical advantage.



Normal area (Barnes): arc swung by forearm pivoting at elbow → used for frequent work

Max. area: arm swung from shoulder → used for occasional work only

Fig 2.7 Normal and maximum working areas in the horizontal plane

4. Micromotion Study, Therbligs and SIMO Chart

- **Micromotion study** (Gilbreth, 1912): detailed study of the very short cycle repetitive work by **filming the operation** (cine camera / video) with a **microchronometer** (a clock reading 1/2000 min) in the field of view – then analysing frame by frame.
- Uses: to analyse and improve fine hand motions of highly repetitive jobs of long run; to train operators; to find the best motion sequence; as a research tool for developing standard data and PMTS.
- Procedure: (1) select job & prepare work-place, (2) film at standard speed (e.g. 1000 frames / min) with the clock, (3) analyse film – break into therbligs, (4) record on SIMO chart, (5) critically examine, (6) develop & film new method, (7) install.
- Limitation: high cost of cine camera / film, time to analyse – justified only for high-volume work. Memo-motion (time-lapse filming at 1 frame per 1/2 – 2 s) is a cheaper tool for long-cycle & multi-person activity.

Therbligs (Gilbreth spelt backwards – "Gilbreth" → "Therblig")

- The basic **elements of hand motion:** 17 (some texts list 18 including Hold / Find). Each has a symbol, letter code and a colour.

Therblig	Code	Description	Type
Search	Sh	Eyes / hand groping to find an object	Ineffective
Select	St	Choosing one article among several	Ineffective
Grasp	G	Closing fingers to hold an object	Effective
Transport empty (Reach)	TE	Moving the empty hand towards an object	Effective
Transport loaded (Move)	TL	Moving an object with the hand	Effective
Hold	H	Retaining the object in a fixed position	Ineffective
Release load	RL	Letting go of the object	Effective
Position	P	Orienting the object during use	Effective / semi
Pre-position	PP	Placing the object ready for the next operation	Effective
Inspect	I	Examining the object for quality	Ineffective / necessary
Assemble	A	Joining two parts together	Effective (productive)
Disassemble	DA	Separating parts	Effective (productive)
Use	U	Using a tool / device for the intended purpose	Effective (productive)
Unavoidable delay	UD	Delay beyond the worker's control	Ineffective
Avoidable delay	AD	Delay within the worker's control	Ineffective
Plan	Pn	Mental process – deciding the next action	Ineffective
Rest (to overcome fatigue)	R	Pause needed to recover from fatigue	Ineffective (needed)
Find	F	Eyes locating an object (end of Search)	Ineffective

Note: Aim: eliminate / shorten ineffective therbligs (Sh, St, H, P, PP, I, Pn, AD, UD) using the principles of motion economy; keep the effective ones (TE, G, TL, A, U, RL) short.

SIMO Chart (Simultaneous Motion Cycle Chart)

- A **two-handed chart in therbligs,** drawn on a common time scale (in **winks:** 1 wink = 1/2000 min = 0.0005 min) from the micromotion film. Shows both hands' therbligs side by side with their durations (in colour codes) – for exact analysis of short repetitive cycles.

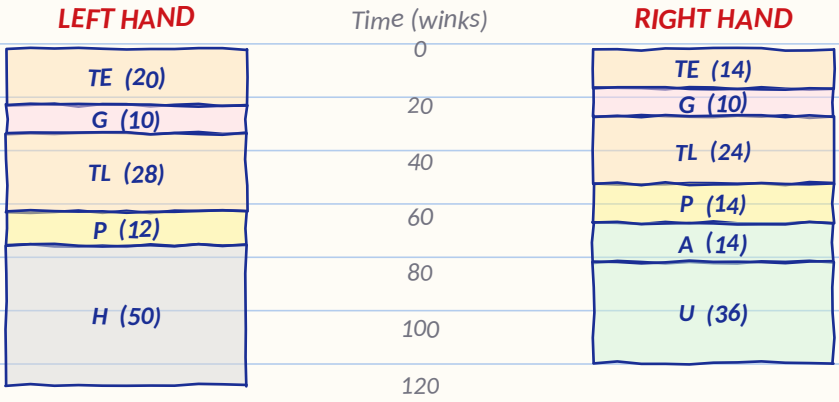
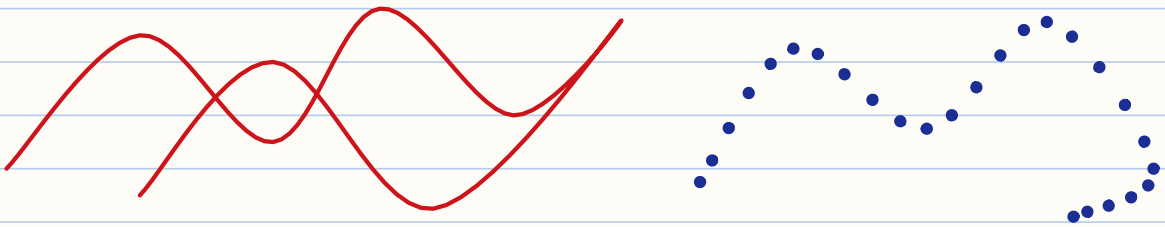


Fig 2.8 SIMO chart (schematic): note H (hold) of LH during RH working → provide a fixture

5. Special Charts – Cyclegraph and Chronocyclegraph

- **Cyclegraph** (Gilbreth, 1917): a record of the **path of a motion** – a small light bulb is fixed to the worker's finger / hand and a still camera takes a long-exposure (time) photograph in a dark room → a continuous bright line shows the path of motion (design of simpler paths, training).
- **Chronocyclegraph:** same but the light is **interrupted at a known rate** (by a rotating shutter or pulsating lamp) → the path appears as a series of dots / dashes (pear-shaped). Spacing of dots tells **speed, acceleration, deceleration, time** of the motion – dots close = slow; far = fast.



Cyclegraph (continuous trace) Chronocyclegraph (dots = time)

Fig 2.9 Cyclegraph and Chronocyclegraph (schematic)

- Applications: to find the ideal motion path, to compare operators, to design work-place layout and training; to study the motions of skilled operators, also in sports and rehabilitation.
- Memo-motion: time-lapse film, used for gang / multi-man activity; micromotion: high-speed fine analysis.

Development, Definition and Installation of the improved method

- **DEVELOP:** apply ECRS and motion economy; prepare alternatives; compare on cost, safety, quality, flexibility; test (pilot trial); select the best.
- **DEFINE:** write a standard-practice sheet (operation, tools, work-place sketch, sequence of motions, standard time); get the approval of management.
- **INSTALL:** (i) get management approval, (ii) win acceptance of supervisors & workers, (iii) train operators step by step, (iv) arrange tools, material, layout, (v) trial run and follow-up, (vi) start full production.
- **MAINTAIN:** audit regularly; settle complaints; update the standard when changes occur; reward people for suggestions.

6. Work Measurement

Definition (ILO)

Work measurement is the application of techniques designed to establish the time for a **qualified worker** to carry out a specified job at a **defined level of performance.**

Objectives / uses

- To fix standard time for output planning, production scheduling, machine loading, delivery dates.
- To estimate manpower and equipment; to balance assembly lines and teams; cost estimation and standard costs.
- To compare alternative methods (after method study); to measure ineffective time and efficiency; to set incentive wages and control labour cost.

Techniques of work measurement

Technique	Brief idea	Best for
Time study (stop-watch)	Direct timing of elements with performance rating	Repetitive short-cycle jobs
Synthesis (standard data)	Build time from stored element times of earlier studies	Jobs with common elements, new jobs
PMTS (MTM, WF, MOST)	Standard times for basic motions from tables	Planning a method before it exists
Work sampling / activity sampling	Random observations of activity → % of time	Long cycle, non-repetitive, many workers
Analytical estimating	Estimate element times by an experienced person	Maintenance, long jobs

Technique	Brief idea	Best for
Estimating / historical data	Past records, rough judgement	Quick, low-accuracy

7. Work Sampling (Activity Sampling / Ratio-Delay Study)

- Developed by L.H.C. **Tippett** (UK textile mills, 1930s). A large number of **instantaneous observations** taken at random intervals of a group of machines / workers; each observation records the activity (working / idle / type of delay) – the fraction of observations ≈ fraction of time.

Need / when to use

- Long-cycle, irregular, non-repetitive or many-person (gang) work; determination of **machine / worker utilisation** and the delay %.
- To find allowances (fatigue, personal, delay); to set the standard time for non-repetitive jobs; to estimate staff requirement (e.g. in offices, maintenance).
- Advantages: observer need not stay continuously; no stop watch; operator not nervous; one observer can study many subjects; can be interrupted & resumed; cheap. Disadvantage: does not give detailed element time; not suitable for short-cycle jobs; observer must be unbiased.

Theory – confidence level & sample size

- Based on the **binomial distribution** (approx. normal for large n). If p = fraction of observations in the state (e.g. idle), n = number of observations, the std. deviation of p is $\sigma_p = \sqrt{p(1-p)/n}$.

Confidence level	Z value (± z σ)	Meaning
68.27 %	1	true p within p ± 1σ
90 %	1.645	p ± 1.645 σ
95 % (most used)	1.96 ≈ 2	p ± 2σ (practical value)
99 %	2.576	p ± 2.58 σ
99.73 %	3	p ± 3σ

Sample size formulae (95 % confidence, z = 2)

Absolute error: $n = 4p(1-p)/L^2$ where L = absolute accuracy (e.g. ±0.05 for ±5 %)

Relative error: $n = 4(1-p)/(S^2 p)$ where S = relative accuracy (e.g. 0.10 for ±10 %)

General: $n = z^2 p(1-p)/L^2$; Error limit $L = z\sqrt{p(1-p)/n}$

Procedure for conducting a work-sampling study

1. Define the objective & activities to be recorded (working, idle, delay-A, delay-B …); decide the confidence level and accuracy.
2. Make a pilot study (50–100 observations) to get a preliminary estimate of p; then compute n.
3. Decide number of observers, days and observations per round → n = (observations per round) × (rounds per day) × (days).
4. Fix **random observation times** using a random-number table / computer (so that observations do not synchronise with the work pattern).
5. Design the observation sheet, brief the workers; make the tours at the random times and record the state at the instant of sight.
6. Compute p = (number of observations in the activity / total observations); check the accuracy with control limits $p \pm 2\sigma$; if outside → take more observations.
7. Report: % utilisation, % delay, allowances; standard time if required.

Standard time from work sampling

Total time observed = T (min); Working % = P; Units produced = N; Rating = R %; Allowance = A %.

Normal time / unit = (T × P × R) / N; Standard time / unit = Normal time × (1 + A)

Problems

P1. A pilot study of 100 observations on a machine shows it idle in 20 observations. Find the number of observations for 95 % confidence and (a) ±5 % absolute accuracy; (b) ±10 % relative accuracy.

Sol: p = 0.20. (a) $n = 4 \times 0.2 \times 0.8 / (0.05)^2 = 0.64 / 0.0025 =$ **256 observations.** (b) $n = 4 \times (1 - 0.2) / (0.10^2 \times 0.2) = 3.2 / 0.002 =$ **1600 observations.**

P2. In a study of 400 observations, a machine was found working in 320 and idle in 80. Find the utilisation and the accuracy at 95 % confidence.

Sol: p(working) = 320/400 = 0.80. $\sigma = \sqrt{(0.8 \times 0.2 / 400)} = 0.02$; 95 % limit = $\pm 2\sigma = \pm 0.04$ → utilisation = **80 % ± 4 %** (76 %–84 %).

P3. Work-sampling data for an operator (8 h day, 480 min): 400 random observations → 80 idle. In the day 120 units were produced. Rating 110 %, allowance 15 %. Find the standard time per unit.

Sol: working % = 320/400 = 80 %. Working time = 480 × 0.80 = 384 min. Normal time = 384 × 1.10 / 120 = **3.52 min.** Standard time = 3.52 × 1.15 = **4.05 min / unit.**

P4. A study of 6 machines is to be made for 5 days with a 95 % confidence, ±5 % accuracy, assuming p = 0.25. How many rounds per day?

Sol: $n = 4 \times 0.25 \times 0.75 / 0.0025 = 300$ observations. Each round = 6 observations → rounds = 300 / 6 = 50 → per day = 50 / 5 = **10 rounds / day.**

8. Quick Revision & Probable Questions

Memory hooks

- Method study: SELECT-RECORD-EXAMINE-DEVELOP-DEFINE-INSTALL-MAINTAIN; ECRS; 5W+1H questioning.
- Symbols: O (operation) T (transport) I (inspection) D (delay) S (storage).
- Barnes: Body – Work place – Tools. Classes of movement: 1 finger … 5 body.
- Wink = 1/2000 min; Cyclegraph = path; Chronocyclegraph = path + speed.
- Work sampling: $n = 4p(1-p)/L^2$ (95 %).

1. Define method study. Explain its objectives and the steps in the procedure.
2. Draw the symbols used in process charts. Differentiate the outline process chart and the flow process chart.
3. Explain flow diagram, string diagram and travel chart with sketches.
4. What is a multiple activity chart? How many machines can one operator attend?
5. State the principles of motion economy under the three groups. Classify the movements.
6. What are therbligs? Explain micromotion study and the SIMO chart.
7. Explain cyclegraph and chronocyclegraph. How is the improved method defined and installed?
8. Define work measurement; list its techniques. What is work sampling? Derive the sample size formula and solve numericals.

UNIT – III

Time Study • Rating • Allowances • Standard Time • PMTS

Syllabus (as given)

Time study – definition, time study equipment, selection of job, steps in time study. Breaking jobs into elements, recording information. Rating: systems of rating, standard rating, standard performance, scales of rating. Allowances: standard time determination, predetermined motion time study (PMTS), factors affecting rate of working, problems on allowances.

1. Time Study – Definition

Definition

Time study is a work-measurement technique for recording the times and rates of working for the elements of a specified job carried out under specified conditions, and for analysing the data so as to obtain the time necessary to carry out the job at a defined level of performance.

- Developed by F.W. **Taylor** (1881, Midvale Steel); the stop-watch is the basic tool, hence also called "stop-watch time study".
- Result: **Standard time** = the time a qualified, trained worker needs to do the job at standard performance, working in the specified method, including allowances for fatigue, personal needs and delays.

Objectives / uses

- To fix standard time for (i) production planning & scheduling, (ii) labour cost & product costing, (iii) wage-incentive plans, (iv) manpower & machine requirements, (v) assembly-line balancing, (vi) comparing methods, (vii) tendering / delivery dates.

Requirements for a good time study

- The method must be **standardised** first (method study before work measurement). The operator should be **qualified,** properly trained and willing; work-place conditions should be normal.
- Suitable for repetitive, short-cycle, manual work with enough volume to justify the cost; the work-study man must be trained and impartial; workers / union informed.

2. Time Study Equipment

1. **Stop-watch:** (a) ordinary (decimal-minute: 1 division = 0.01 min; 100 divisions per rev.); (b) split-hand / fly-back type; (c) decimal-hour watch (1 division = 0.0001 h); (d) electronic digital stopwatch / data recorder (0.01 s resolution, memory, can download).
2. **Time study board:** light plywood / plastic with a clip, holds the observation sheet and mounts the stop-watch (at elbow height).
3. **Time study observation sheets / forms:** for general information, element description, readings, rating, calculations.
4. Other: stop-clock, **video camera** / movie camera (for micromotion), tachometer / spring balance (to measure speed, force), steel tape, pencils, calculator / laptop, **Time-study software**, stopwatch counters.



crown (start / stop / reset)

Decimal-minute stop-watch

- 100 divisions = 1 min (1 div = 0.01 min)
- small dial counts 30 min
- fly-back: press = return to zero & restart
- mounted on the board, held at elbow level

Fig 3.1 Decimal-minute stop-watch (sketch)

Note: A watch should be sturdy, accurate, checked / calibrated periodically; electronic timers are more accurate and give less reading error.

3. Selection of the Job

- Reasons for study: new job / new product; change of method, material or tools; complaints about the standard (workers or union); a bottleneck or high-cost operation; poor machine utilisation; a basis for incentives.
- Selection by: **economic** importance (high labour content, large volumes), **technical** (quality, repetitiveness), **human** factors (fatigue, complaint). Choose a **qualified & average operator** (neither the fastest nor slowest).
- Inform the supervisor, union representative and operator; explain the purpose; stand where you can see the work without disturbing the operator.

4. Steps in Time Study (ILO – 8 steps)

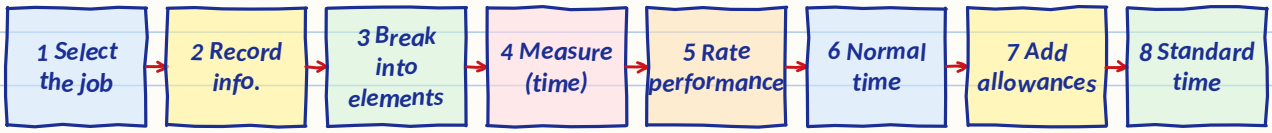


Fig 3.2 Steps in a stop-watch time study

1. **Select** the job to be studied (and the operator).
2. **Obtain & record** all information about the job, operator, tools, material, machine, conditions, drawing, layout.
3. **Divide the job into elements** (breaking up) and describe them with start / end break-points.
4. **Measure the time** taken by the operator for each element with the stop-watch; take a sufficient number of cycles (N).
5. **Assess the effective speed** of working of the operator relative to the standard (rating).
6. **Convert the observed times to normal (basic) times** : NT = OT × (Rating / 100).
7. **Determine the allowances** (relaxation, contingency, special) and add them.
8. **Compute the standard time** of the job = normal time + allowances.

5. Breaking the Job into Elements

- **Element** = a distinct part of a specified job, selected for convenience of observation, measurement and analysis. A job is divided so that: (i) the rating can be applied per element; (ii) elements with different characteristics are separated; (iii) the standard data library can be built; (iv) abnormal / foreign elements can be excluded.

Rules / guidelines for breaking into elements

- Elements should be **short enough** to be timed accurately but not so short that readings are inaccurate (minimum ≈ 0.04 min (2.4 s) for a watch reader; 0.1 min to be safe).
- Separate **manual** elements from **machine** (power-fed) elements; separate **constant** elements (same time every cycle) from **variable** elements (time depends on size, weight, length etc.).
- **Regular** elements in every cycle; **irregular / occasional** elements (e.g. sharpening a tool) recorded separately and spread over the cycles.
- **Foreign (extraneous) elements** – unavoidable delay or not part of the job (talking, dropped tool) – are noted and excluded from the time.

- Choose break-points that are clear: a sound, a visual change (e.g. "part clicks into fixture") – the end of one element = start of the next.
- **Governing element** = in a cycle with simultaneous elements, the longest (e.g. a machine run) governs the time.

Type of element	Meaning	Example (turning a shaft)
Repetitive / regular	Occurs every cycle	Pick up bar, clamp in chuck
Occasional	Occurs at intervals	Re-sharpen tool each 50 pieces
Constant	Same time every cycle	Start the machine
Variable	Time changes with the work	Turn length varies with diameter
Manual / Machine	Done by worker / by machine power	Load (manual) / feed (machine)
Foreign	Not part of the job	Operator wipes his face

6. Recording Information (Time Study Form)

- Record on the **top of the observation sheet:** study number, date, name and number of the operator, department, observer; description of the operation, part name / number, drawing, material, machine & tool (type, number, speed, feed), work-place layout sketch, conditions (light, temperature), start & end time of the study. All details that affect time, so that a future person can reproduce the study.
- **Methods of timing** using the stop-watch:

(a) **Continuous timing** – the watch runs continuously from the start; time is noted at the end of each element (reading R); element time = difference of successive readings. Foreign elements shown; no time lost; used for short elements.

(b) **Repetitive / snap-back timing** – the hands are returned to zero at the end of each element; the reading = element time. Simple, but time of the snap-back is lost, errors for short elements.

(c) **Accumulative timing** – two watches mounted on the board connected by a lever: when one runs the other is stopped; the second is read while the first runs. Direct element times without subtraction.

Note: Use a sufficient number of readings (6–10 cycles to begin with); discard abnormal / foreign readings.

Elem.	Description	Cycle 1	2	3	4	5	Avg (OT)	R%	NT
A	Pick up part	0.20	0.22	0.19	0.20	0.21	0.204	110	0.224
B	Drill hole (machine)	1.50	1.52	1.50	1.49	1.50	1.502	100	1.502

Elem.	Description	Cycle 1	2	3	4	5	Avg (OT)	R%	NT
C	Remove, place in bin	0.30	0.28	0.32	0.30	0.30	0.300	90	0.270

Note: Sample record: OT = observed time (average) in min; R = rating; NT = OT × R/100. Rating is not applied to the machine-controlled element (B).

Number of cycles to be timed

- Depends on cycle time & accuracy; ILO / General Electric table: cycle 0.10 min → 200 cycles; 0.25 → 100; 0.50 → 60; 0.75 → 40; 1.0 → 30; 2.0 → 20; 5.0 → 10; 10 → 8; 20 → 5; 40 → 3; over 40 → 2.
- Statistical formula (95 % confidence, ±5 % accuracy): $N = (40 \sqrt{(n \Sigma x^2 - (\Sigma x)^2) / \Sigma x)^2}$ n = number of readings in the trial, x = each reading.

7. Rating (Performance Rating / Levelling)

Definition

Rating is the assessment of the worker's rate of working relative to the observer's concept of the rate corresponding to **standard performance.**

Normal time = Observed time × (Rating / 100) (Rating factor R = observed rate / standard rate)

- Why rating? The operator observed may work faster or slower than the standard. Rating converts the observed time into the time a **normal, qualified worker** would take.

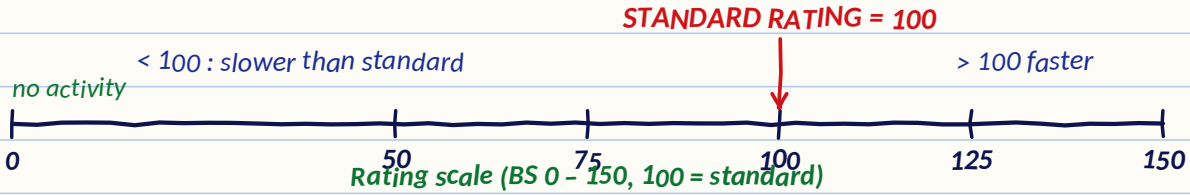
Standard performance and standard rating

- **Standard performance** (BS 3138): the rate of output which qualified workers will naturally achieve without over-exertion as an average over the working day, provided they know and adhere to the specified method and provided they are motivated to apply themselves to their work. It is denoted as 100 on the BS 0–100 scale.
- **Standard rating** = the rating corresponding to the standard performance (100 on the BS scale; 60 on the 60–80 scale; 75 on the 75–100 scale).
- Bench-marks: walking 4 miles / hour (≈ 6.4 km/h) without load on level ground = standard (100); dealing a pack of 52 playing cards into four piles in 0.5 minute = standard.

Scales of rating

Scale	Standard (normal)	Incentive (≈ +33 %)	Remarks
0 – 100 (British Standard)	100	133	0 = no activity; 100 = standard; > 100 faster
60 – 80 (American)	60	80	80 = incentive pace
75 – 100	75	100	100 = incentive pace
100 – 133	100	133	Same as BS scale

Note: Example: observed rating 90 on the BS scale = 90 % of standard speed → R = 0.90.



Systems of rating

1. **Speed rating** (pace rating): observer assesses only the speed / tempo of movement against the standard concept (single factor) – R = observed pace / standard pace. Simple, quick; depends on observer's judgement and training (use training films).
2. **Westinghouse system (levelling):** four factors – **Skill, Effort, Conditions, Consistency;** each graded (A to F) and a numeric ± value from the table; Rating factor = 1 + algebraic sum.
3. **Synthetic rating** (Morrow): the observed time of a few elements is compared with the PMTS (e.g. MTM) time; R = P / A (P = predetermined time, A = actual observed time); average R is applied to the other elements.
4. **Objective rating** (Mundel): two steps – (i) rate the speed against a standard reference (pace only); (ii) add a **job-difficulty adjustment** (amount of body used, foot pedals, bi-manual, eye-hand coordination, handling / sensory requirements, weight handled) from tables → Rating = Pace rating × Secondary adjustment factor.
5. **Performance rating (physical)** / skill & effort rating (Lowry et al.) – and "pace rating" with training films by the Society for Advancement of Management (SAM).

Westinghouse rating table

Grade	Skill	Effort	Conditions	Consistency
A1 / A	Super skill +0.15	Excessive +0.13	Ideal +0.06	Perfect +0.04
A2	+0.13	+0.12		
B1 / B	Excellent +0.11	Excellent +0.10	Excellent +0.04	Excellent +0.03

Grade	Skill	Effort	Conditions	Consistency
B2	+0.08	+0.08		
C1 / C	Good +0.06	Good +0.05	Good +0.02	Good +0.01
C2	+0.03	+0.02		
D	Average 0.00	Average 0.00	Average 0.00	Average 0.00
E1 / E	Fair –0.05	Fair –0.04	Fair –0.03	Fair –0.02
E2	–0.10	–0.08		
F1 / F	Poor –0.16	Poor –0.12	Poor –0.07	Poor –0.04
F2	–0.22	–0.17		

- **Skill** = proficiency in following a given method; **Effort** = willingness to work (speed with which skill is applied); **Conditions** = working conditions affecting the operator (not the operation) – light, temperature, noise; **Consistency** = variation in repeated times (fine, rating when elements repeat).
- Rating factor R = 1 + (Skill + Effort + Conditions + Consistency). Rating % = R × 100.

8. Allowances and Standard Time Determination

- Normal time is the time for a worker working at standard rating with **no allowance for rest, personal needs or delay.** A worker cannot work continuously; allowances are added to get a time which can be achieved in actual conditions.

Allowance	Purpose	Typical value
Personal needs (relaxation)	Drinking water, toilet, washing, tea; depends on climate and work	5 % (men), 7 % (women)
Basic fatigue	Energy consumed + monotony while working under ideal conditions	4 % of normal time
Variable fatigue	Special: standing / awkward posture, heavy force, poor light, heat, noise, close attention, monotony, mental strain	2 – 20 % (table / ILO)
Contingency	Small unavoidable delays, irregular elements not timed, e.g. tool change, inspection	≤ 5 %
Delay (unavoidable)	Machine breakdown, power failure, waiting for materials / instructions	by study / sampling
Special / policy	Training, setting-up, interference (multi-machine), tool & process allowances, learning, management policy	by policy

Standard time determination

Observed time (OT) → **Normal time NT = OT × R / 100** → add allowances → **Standard time ST = NT + (Allowances) = NT × (1 + A)** (A = allowance as a fraction of normal time).

When allowance is given as a % of the **total (shift) time** : **ST = NT / (1 – A)** (e.g. 15 % → ST = NT / 0.85).

Standard output per day = shift time / ST; Performance index (efficiency) = (standard time × output) / (time on the job) × 100.

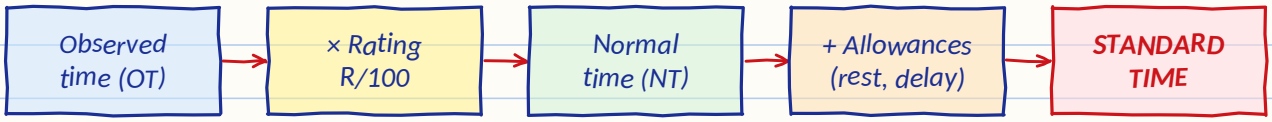


Fig 3.3 Build-up of standard time

9. Predetermined Motion Time Systems (PMTS)

- Definition: a work-measurement technique in which **times established for basic human motions** (classified by the nature of the motion and the conditions under which it is made) are used to build up the time for a job at the defined level of performance.
- Principle: break the job into basic motions (reach, move, turn, grasp, position, disengage, release, eye travel, body motions); look up the time for each from tables (depends on distance, weight, case, difficulty); add them. No stop-watch or rating needed, so it can be applied **before the job exists** (during the design of the method).

System	Developer / year	Time unit	Remarks
Work-Factor (WF)	Quick, Duncan & Malcolm (1938)	1 RWF = 0.0001 min	Work factors for body member, distance, manual control, weight
Methods-Time Measurement (MTM-1)	Maynard, Stegemerten & Schwab (1948)	1 TMU = 0.00001 h = 0.0006 min = 0.036 s	Most widely used; MTM-2, MTM-3 (faster, less detailed)
MOST (Maynard Operation Sequence Tech.)	Zandin (1967)	TMU (index × 10)	Sequence models: General move (ABGABPA), Controlled move, Tool use

System	Developer / year	Time unit	Remarks
BMT (Basic Motion Time Study)	Brisch / Mackie (1950s)	1 BMU = 0.0001 min	Basic motions with time tables
DMT / Dimensional Motion Times	Wygant & Sellie	—	Dimension-based motion times

- MTM-1 basic motions: **Reach (R), Move (M), Turn (T), Apply pressure (AP), Grasp (G), Position (P), Release (RL), Disengage (D),** Eye travel (ET) & Eye focus (EF), body, leg and foot motions (walk, bend, kneel, sit).
- Conversions (MTM): 1 TMU = 0.036 s; 1 min = 1666.7 TMU; 1 h = 100 000 TMU.

Merits of PMTS	Demerits of PMTS
No stop watch or rating – objective & consistent	Needs trained analyst; data tables are costly (licence)
Method can be assessed before production starts	Detailed analysis takes long time; not for long-cycle / mental work
Helps in method improvement, plant layout, line balancing	Not accepted by some workers; applicable to manual work only
Gives accurate standard data library	Limited to the motions in the tables

10. Factors Affecting the Rate of Working

- Rate of working (speed) of an operator is not constant; it depends on:

Factor	Examples
Operator	Skill, training, experience, health, age, physical fitness, attitude, motivation, mood, personal problems
Job / method	Difficulty, accuracy needed, weight, work-place layout, tools & fixtures, motion economy of the method
Material / machine	Quality of material, machine condition, speeds & feeds, tooling
Environment	Temperature, humidity, lighting, noise, ventilation, vibration, cleanliness
Organisation / management	Supervision, incentive scheme, wage system, job security, group relations, work load, time of day, shift, fatigue pattern
Time-study situation	Awareness of being observed, observer's attitude, union stand, fear of rate cutting

Note: Management must provide an environment where the worker can attain standard performance without excessive fatigue.

11. Standard Data, Merits & Limitations of Time Study

- **Standard data** (synthesis): element times (constant & variable) collected from many earlier time studies, tabulated or as formulas / curves (e.g. time = a + b × length). New jobs are built up by adding the standard element times – no new stop-watch study; consistent standards, quick estimates, lower cost.
- Example: drilling time T = (L / (f × N)) where L = depth + approach (mm), f = feed (mm/rev), N = rpm; handling time from standard data tables per weight / distance.

Merits of time study	Limitations of time study
Gives direct, observed, and accurate time for short repetitive work	Needs trained observers; rating is subjective (±10 % error)
Basis for fair standards, planning, cost, incentive wages	Not suited to long-cycle, non-repetitive & mental work
Detects delays & method faults during the study	Workers may feel nervous / act abnormally when watched
Simple equipment; widely accepted by industry	Cost & time of repeated studies; becomes obsolete when method changes

Point	Time study	Work sampling
Observation	Continuous, with stop watch	Random instantaneous glimpses
Job type	Short, repetitive, manual	Long, irregular, group work
Rating	Needed	Needed only if standard time is desired
Output	Element time, standard time	% of time in each activity, delays, allowances
Operator effect	Observed continuously – may be disturbed	Less disturbing

Precautions while rating and timing

- Rate each element **while it is being performed** (not after); rate only the speed of working, not the operator's skill unless the system requires it; ignore the effect of observer's personal liking.
- Do not rate the machine-controlled elements; use training films / many practice sessions to calibrate the rating (check against other observers: rating accuracy ±5 %).

- Use the **average of the selected readings** after rejecting abnormal values (> 2 × the usual); average the ratings of the elements with different rating factors separately.

Learning curve and fatigue (side notes)

- New operators need time to reach standard output – training allowance / learning curve (the time per unit falls by a constant % each time output is doubled: 80 % curve is typical).
- Fatigue is reduced by good method (motion economy), proper rest breaks, good environment, rotation of jobs; the relaxation allowance recovers the energy expended.

12. Problems on Time Study, Rating and Allowances

P1. The average observed time for an element is 0.50 min. The worker is rated 110 %. Allowances are 15 % of normal time. Find the standard time and the output for an 8-hour shift.

Sol: NT = 0.50 × 1.10 = 0.55 min. ST = 0.55 × (1 + 0.15) = **0.6325 min.** Output = 480 / 0.6325 = **759 pieces / shift** (≈ 758.8).

P2. Same data, but the 15 % allowance is given as % of the total (shift) time. Find the standard time.

Sol: ST = NT / (1 – 0.15) = 0.55 / 0.85 = **0.647 min.**

P3. A time study gave average times & ratings: (1) pick-up 0.20 min, 110 %; (2) machining (machine-controlled) 1.50 min, 100 %; (3) remove & place 0.30 min, 90 %; (4) inspect 0.25 min, 120 %. Allowances 15 %. Find the standard time and output per 8 h.

Sol: NT = 0.20×1.10 + 1.50×1.00 + 0.30×0.90 + 0.25×1.20 = 0.22 + 1.50 + 0.27 + 0.30 = 2.29 min. ST = 2.29 × 1.15 = **2.63 min.** Output = 480 / 2.63 = **182 pieces.**

P4. Westinghouse rating: skill – good (C1), effort – good (C2), conditions – average (D), consistency – excellent (B). Observed time 0.90 min. Find the normal time.

Sol: Skill +0.06; Effort +0.02; Conditions 0; Consistency +0.03 → sum = +0.11. R = 1.11. NT = 0.90 × 1.11 = **0.999 ≈ 1.0 min.**

P5. Ten stop-watch readings (min): 0.5, 0.6, 0.5, 0.7, 0.6, 0.5, 0.6, 0.5, 0.7, 0.6. Find the number of cycles to be timed (95 % confidence, ±5 %).

Sol: $\Sigma x = 5.8$; $\Sigma x^2 = 3.42$; $n = 10 \rightarrow n\Sigma x^2 - (\Sigma x)^2 = 34.2 - 33.64 = 0.56$; $\sqrt{0.56} = 0.748$. $N = (40 \times 0.748 / 5.8)^2 = (5.16)^2 = 26.6 \approx$ **27 cycles.**

P6. Observed average cycle 2.0 min; rating 85 %; allowances: personal 5 %, fatigue 5 %, delay 3 %. Find standard time.

Sol: NT = 2.0 × 0.85 = 1.70 min; total allowance = 13 %; ST = 1.70 × 1.13 = **1.92 min.**

P7. A job analysed by MTM needs 250 TMU. Find the time in seconds and minutes; with 10 % allowance find standard time.

Sol: Time = 250 × 0.036 = 9.0 s = 0.15 min. ST = 9.0 × 1.10 = **9.9 s = 0.165 min.**

P8. An operator makes 300 pieces in 8 h; standard time of the job = 1.4 min. Find the performance index.

Sol: Standard minutes produced = 300 × 1.4 = 420; available time = 480 min.

Performance = 420 / 480 × 100 = **87.5 %.**

P9. An operator attends 3 identical machines. Loading / unloading time per machine = 1.2 min, machine running time = 4.0 min, walking time between machines = 0.2 min. Find the machines per operator (ideal) and the cycle time for 3 machines.

Sol: $n = (l + m)/(l + w) = (1.2 + 4.0)/(1.2 + 0.2) = 5.2 / 1.4 = 3.7$ → check $n = 3$ and $n = 4$.

For $n = 3$: machine-cycle = $l + m = 5.2$ min; operator work = $3 \times (1.2 + 0.2) = 4.2$ min < 5.2 → cycle time = **5.2 min per 3 pieces** (machine-governed); operator idle = 1.0 min per cycle.

P10. A job has a normal time of 8 min. Allowances: personal 5 %, fatigue 4 %, special 3 %, all as % of the total (shift) time. Find ST and the output in 8 h.

Sol: total allowance = 12 % → ST = 8 / (1 – 0.12) = **9.09 min.** Output = 480 / 9.09 = **52.8 ≈ 52 pieces / shift.**

13. Quick Revision & Probable Questions

Memory hooks

- ST = (OT × R/100) × (1 + A); with shift-based allowance: ST = NT/(1 – A).
- Westinghouse: Skill, Effort, Conditions, Consistency (S E C C).
- Timing: continuous – snap-back – accumulative. Wink / TMU: 1 TMU = 0.036 s.
- Time study steps: Select – Record – Break – Measure – Rate – Normal – Allowances – Standard.

1. Define time study. Explain the equipment used and the steps in a time study.
2. How is a job broken into elements? State the rules. Explain methods of timing.
3. What is rating? Explain standard performance and the scales of rating. Describe the Westinghouse system.
4. List the allowances. How is standard time determined?
5. What is PMTS? Explain MTM / Work Factor / MOST with merits and demerits.
6. Discuss the factors affecting the rate of working.
7. Numericals on rating, allowances, number of cycles and standard time (P1 – P8).

UNIT – IV

Introduction to Ergonomics • Physical • Cognitive • Organizational • Environmental

Syllabus (as given)

Introduction to Ergonomics: human factors and ergonomics, psychology, engineering, biomechanics, industrial design, graphic design, statistics, operations research and anthropometry. Morphology of design and its relationship with cognitive abilities of human beings. Physical ergonomics: human anatomy, anthropometric, physiological and biomechanical characteristics as they relate to physical activity. Cognitive ergonomics: mental processes – perception, memory, reasoning, motor response, mental workload, decision-making. Organizational ergonomics: optimisation of socio-technical systems, organisational structures, policies, processes; communication, work design, design of working times, teamwork, cooperative work, new work programs. Environmental ergonomics: human interaction with the environment – climate, temperature, pressure, vibration, light.

1. Ergonomics – Definition, Origin and Objectives

Definition (IEA – International Ergonomics Association, 2000)

Ergonomics (or human factors) is the scientific discipline concerned with the understanding of interactions among humans and other elements of a system, and the profession that applies theory, principles, data and methods to design in order to optimise human well-being and overall system performance.

- Word origin: Greek **ergon = work** + **nomos = natural laws** → "laws of work". Coined by K.F.H. **Murrell** (UK, 1949); the US term is **Human Factors (Engineering).** Other names: human engineering, biotechnology, engineering psychology.
- Simple definition: **fitting the job (task, tool, environment) to the person** – not the person to the job; "design for human use".

Brief history

- Early roots: Taylor's scientific management and the Gilbreths' motion study (work physiology, fatigue); Hawthorne studies (1924–32, social factors).

- World War II: aircraft cockpits, radar displays and weapons – operator errors due to poor design → **human factors engineering.** Post-war: Ergonomics Research Society (1949, UK); Human Factors Society (1957, USA); IEA (1959).
- Today: product design, computers & mobile devices, healthcare, transportation, safety, manufacturing, sports.

Objectives and benefits

- **Improve safety, health and comfort** – reduce accidents, musculoskeletal disorders (MSDs), fatigue and stress.
- **Improve performance & productivity** – reduce errors, increase speed and quality; reduce absenteeism, training time and cost.
- Improve user satisfaction, acceptability and usability of products and systems; match demands of the task with capabilities and limitations of humans.

2. Human Factors and Ergonomics – a Multidisciplinary Subject

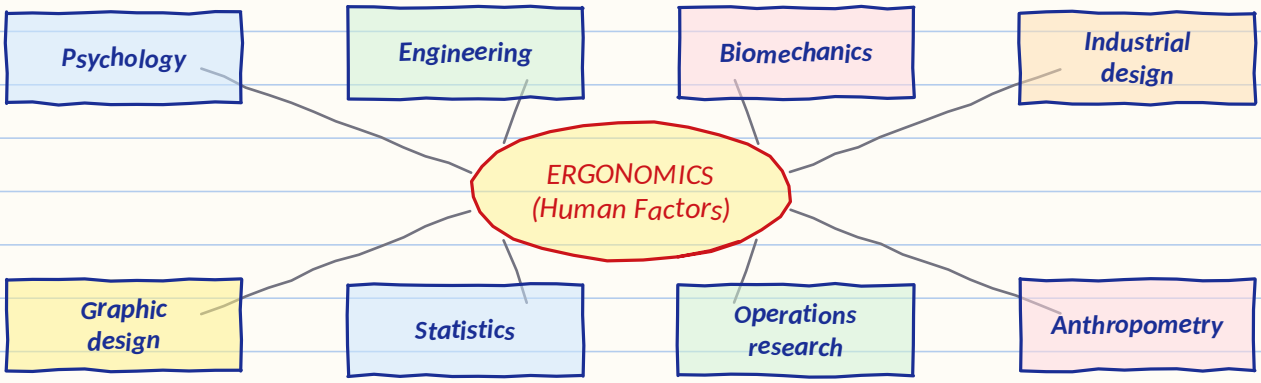


Fig 4.1 Ergonomics draws on many disciplines

Discipline	Contribution to ergonomics
Psychology	Perception, attention, memory, learning, motivation, stress, human error, mental workload, user behaviour
Engineering	Design of machines, controls, work systems, safety, human–machine system analysis, production & maintenance
Biomechanics	Application of mechanics to body: forces on joints & spine, lifting, posture, muscle strength, tool handles
Industrial design	Form, shape, aesthetics and usability of products, tools, furniture; human-centred styling
Graphic design	Visual communication: symbols, layout, colour, typography, displays, labels, user interface

Discipline	Contribution to ergonomics
Statistics	Design of experiments, sampling, percentiles, variability of human data, hypothesis testing
Operations research	Modelling & optimisation of human–machine systems, scheduling shift work, queuing, decision analysis
Anthropometry	Measurement of body dimensions; data for sizes of workspace, seats, tools, clothing; percentiles

- Also: physiology (work capacity, fatigue), anatomy, medicine / occupational health, sociology, architecture, computer science / HCI.

3. Morphology of Design and its Relationship with Cognitive Abilities

- **Morphology** = the study of form and structure. In design, **design morphology** means the systematic study of the form, structure, parts and their arrangement in a product (shape, size, layout, colour, texture, proportion) and how they are generated and combined.
- **Morphological analysis** (Zwicky): break the product into essential functions / parameters, list the possible options for each (a morphological matrix) and combine options to get new design concepts.

Function / parameter	Option 1	Option 2	Option 3
Grip of a hand tool	Cylindrical	Pistol grip	Contoured
Material	Wood	Plastic	Rubber coated
Activation	Push button	Trigger	Lever

- Stages of the design process: (1) problem / user analysis → (2) concept generation (morphology) → (3) embodiment: form, layout, dimensions → (4) detail design & prototype → (5) user testing & evaluation → iterate.

Relationship with cognitive abilities

- A product's form is "read" by the user's mind. If the form matches how humans **perceive, remember and think,** the product is easy to use; if not, errors and frustration occur.

Cognitive ability	Design implication (form / morphology)
Perception (Gestalt: proximity, similarity, closure, continuity)	Group related controls; use similar shape for similar function; clear visual hierarchy

Cognitive ability	Design implication (form / morphology)
Attention (selective, limited)	Make critical information salient: colour, size, position; avoid clutter
Memory (STM 7 ± 2 items; LTM)	Limit choices; chunking; recognition rather than recall; use familiar symbols
Mental models & expectations	Follow population stereotypes (clockwise = increase; red = stop)
Learning	Consistency, feedback, progressive disclosure, intuitive forms
Decision-making & reasoning	Present clear options, show state of system, support error recovery

Norman's design principles

- **Affordance** – the form suggests how it is to be used (a handle affords pulling).
- **Signifiers & visibility** – make possible actions visible.
- **Mapping** – natural spatial relation between controls and effects.
- **Feedback** – immediate indication of the result of an action.
- **Constraints** – form that prevents wrong action (poka-yoke); **consistency** across the system.

- Laws used in design: **Hick's law** – decision (choice reaction) time increases with log of number of choices: $RT = a + b \log_2(n + 1)$. **Fitts' law** – movement time to a target: $MT = a + b \log_2(2D / W)$ (D = distance, W = target width) – large, near targets are quicker.

4. Domains of Ergonomics (IEA)

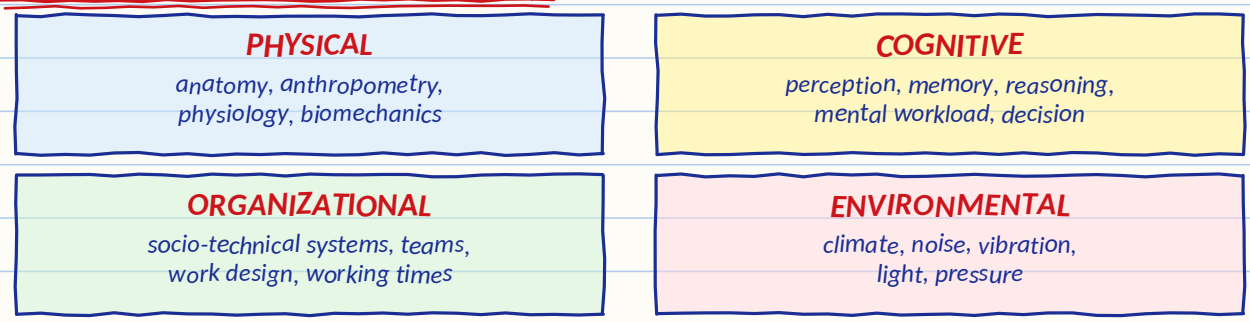


Fig 4.2 Four domains of ergonomics (IEA)

5. Physical Ergonomics

- Concerned with human anatomical, anthropometric, physiological and biomechanical characteristics as they relate to physical activity. Topics: working postures, materials handling, repetitive movements, work-related musculoskeletal disorders (WMSDs), workplace layout, safety and health.

(a) Human anatomy relevant to ergonomics

- **Skeletal system:** 206 bones – support, protect organs, levers for movement; **joints** (hinge – elbow, knee; ball & socket – shoulder, hip; pivot, gliding). The **spine** has 33 vertebrae: cervical (7), thoracic (12), lumbar (5), sacral (5 fused), coccyx (4 fused). Lumbar discs carry the highest load in lifting (L5/S1).
- **Muscular system:** skeletal muscle (voluntary), cardiac and smooth (involuntary). Muscle contraction: isotonic (movement), isometric (static holding – causes fatigue quickly), concentric / eccentric.
- Tendons connect muscle to bone; ligaments connect bone to bone; **nervous system** (CNS, PNS) controls; **circulatory & respiratory systems** supply O2 & remove waste (determine work capacity).
- Static muscular work restricts blood flow – leads to quick fatigue; dynamic work with rhythmic contraction is better.

(b) Anthropometry – body dimensions

- **Anthropometry** = measurement of the human body (size, shape, strength, mobility). Used to decide **dimensions of workplace, seats, controls, tools, clothing.**
- Types: **Static (structural)** – measured in fixed positions (stature, sitting height, arm length); **Dynamic (functional)** – measured during movement (reach envelope, range of joint motion); also strength measurement.

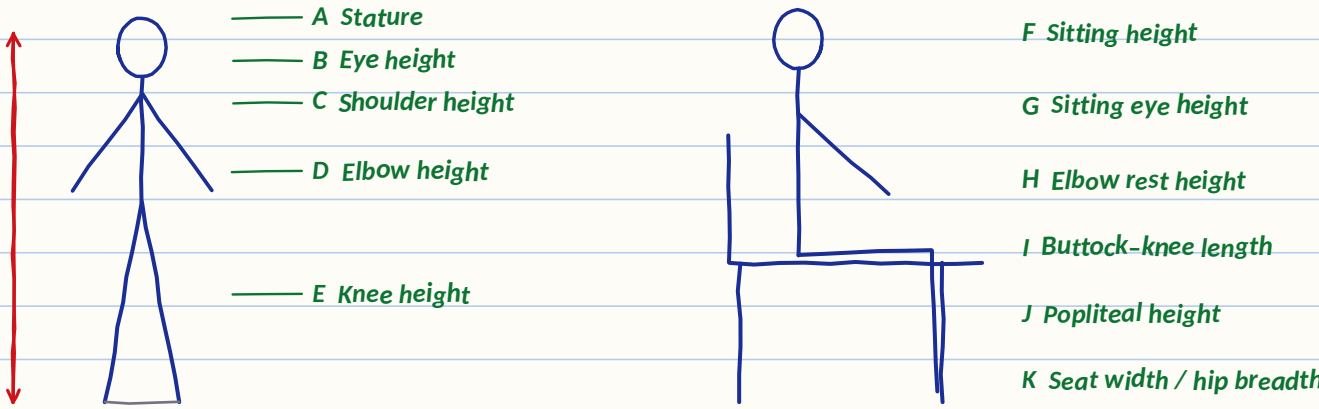
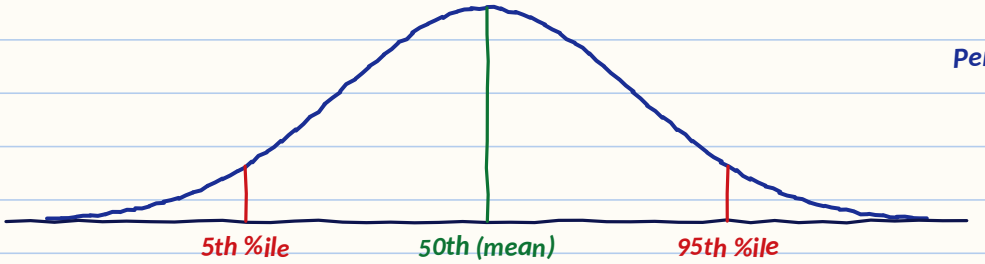


Fig 4.3 Common anthropometric measurements (standing & seated)

- Human data are **variable** – age, sex, race / region, occupation, generation (secular growth). Data are given as **mean, standard deviation and percentiles** (normal distribution).



Percentile value = mean ± z·σ
5th = mean – 1.645 σ
95th = mean + 1.645 σ
99th = mean + 2.33 σ

Fig 4.4 Normal distribution of a body dimension – percentiles

Principles of applying anthropometric data

Design principle	Use	Examples
Design for extremes (limiting dimension)	Min. or max. of population	Door height, clearance, leg room → 95th (max); reach to controls, seat height, force to operate lever → 5th (min)
Design for adjustability	Range 5th – 95th %ile (90 % of users)	Chair height, monitor height, car seat, steering
Design for the average	Where adjustment is not feasible	Counter height in a shop, hand-rails, checkout
Design for the individual	When dimension is for one person	Custom tool handle, prosthetics

Note: Always add allowances for clothing, shoes, posture slump, and use dimensions of the target population (e.g. Indian data).

(c) Physiological characteristics

- **Work physiology:** energy (ATP) supplied by aerobic and anaerobic processes; oxygen consumption (litres/min) and heart rate are measures of workload; 1 litre O_2 ≈ 5 kcal.

Work category	Energy expenditure (kcal / min)	Heart rate (beats / min)
Light	2.0 – 4.9	< 100
Moderate	5.0 – 7.4	100 – 120
Heavy	7.5 – 9.9	120 – 140
Very heavy	10.0 – 12.4	140 – 160
Extremely heavy	≥ 12.5	> 160

- Resting metabolism ≈ 1.5 kcal / min; recommended average workload for an 8-h shift ≈ 5.0 kcal/min (men) and ≈ 4.0 kcal/min (women) – above that rest pauses are needed.

Rest allowance – Murrell's formula

R = T × (W – S) / (W – 1.5) R = rest time (min), T = total work time (min), W = energy expenditure of the work (kcal/min), S = recommended sustainable level (5.0 men / 4.0 women).

Example: W = 7.5 kcal/min, T = 480 min (men): R = 480 × (7.5 – 5.0)/(7.5 – 1.5) = 480 × 2.5 / 6 = **200 min** of rest per shift.

- Fatigue: physical (muscle) and mental; causes – heavy / static load, long hours, monotonous work, poor environment; symptoms – reduced output, errors, discomfort. Remedy – job design, rest pauses, rotation, tools, training.
- Max. aerobic capacity (VO2 max) ≈ 3.0 l/min (men) and 2.0 l/min (women); the safe continuous workload ≈ 33 % of VO2 max for 8 h.

(d) Biomechanical characteristics

- **Biomechanics** = application of mechanics to the body: the body is treated as a system of **levers** (bones), **joints** (fulcrum) and muscles (effort). Important – forces, moments, posture, strength, tolerance of tissues.
- Principles: keep loads close to the body (short moment arm); avoid twisting & bending of the trunk; use legs not back; neutral posture of the joints; avoid static load; use mechanical aids; reduce repetition & force; reduce vibration.
- **WMSDs / RSI:** carpal tunnel syndrome, tendinitis, tennis elbow, low-back pain, neck & shoulder strain – risk factors: force, repetition, awkward posture, static load, duration, vibration, cold.
- Posture assessment tools: **OWAS, RULA** (Rapid Upper Limb Assessment), **REBA** (Rapid Entire Body Assessment); lifting tool: **NIOSH Lifting Equation.**

NIOSH lifting equation (metric)

RWL = LC × HM × VM × DM × AM × FM × CM (Recommended Weight Limit, kg)

LC = 23 kg (load constant); HM = 25/H (H = horizontal distance, cm); VM = 1 – 0.003 | V – 75|; DM = 0.82 + 4.5/D; AM = 1 – 0.0032 A (asymmetry); FM = frequency, CM = coupling.

Lifting Index LI = Load / RWL; LI > 1 → risk of injury – redesign task.

Example: H = 40, V = 60, D = 40, A = 0, FM = CM = 1 → HM = 0.625, VM = 0.955, DM = 0.9325 → RWL = 23 × 0.625 × 0.955 × 0.9325 ≈ **12.8 kg;** for a 15-kg load LI = 15/12.8 = 1.17 > 1.

6. Cognitive Ergonomics

- Concerned with mental processes – **perception, memory, reasoning, motor response** – as they affect interactions among humans and other elements of a system. Topics: mental workload, decision-making, skilled performance, human–computer interaction, human reliability, work stress and training.

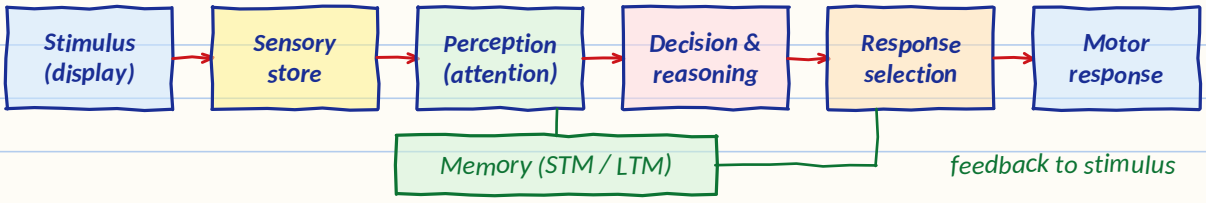
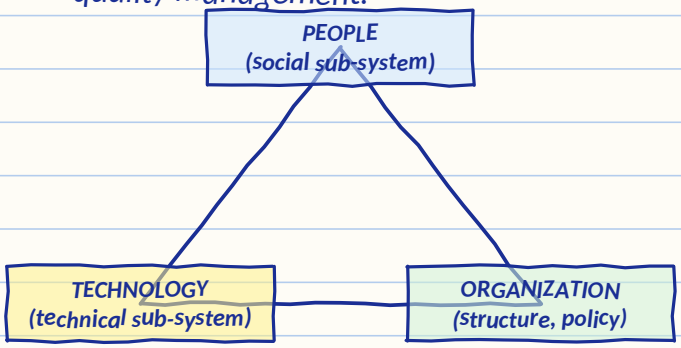


Fig 4.5 Human information-processing model

- **Perception:** sensing (vision, hearing, touch) and interpretation of stimuli; depends on contrast, size, expectation; illusions & limitations (absolute / difference thresholds – Weber's law $\Delta I / I$ = constant).
- **Attention:** selective, divided, sustained (vigilance). Vigilance falls after ≈ 30 min on monitoring tasks.
- **Memory:** sensory memory (≈ 1–4 s), short-term / working memory (≈ 7 ± 2 chunks, 15–30 s without rehearsal), long-term memory (large, permanent). Design: reduce memory load, use recognition.
- **Reasoning & decision-making:** deductive / inductive, heuristics and biases; skill-, rule- and knowledge-based behaviour (Rasmussen). Time pressure, uncertainty and information overload degrade decisions.
- **Motor response:** reaction time (simple visual ≈ 0.2–0.25 s, auditory ≈ 0.15–0.20 s, choice RT longer – Hick's law); speed–accuracy trade-off (Fitts).
- **Mental workload:** the demand placed on the human information-processing capacity. Measures: (i) primary / secondary task performance, (ii) **subjective** – NASA-TLX, SWAT, Cooper–Harper; (iii) **physiological** – heart rate variability, EEG, pupil diameter. Both under-load and overload are harmful.
- **Human error** (Reason): **slips** & **lapses** (right intention, wrong execution / memory failure), **mistakes** (wrong plan – rule / knowledge based), **violations.** Prevent via design: constraints, feedback, redundancy, checklists, training.
- **Situation awareness** (Endsley): perception of elements → comprehension → projection of future states. Important for pilots, control-room operators, drivers.

7. Organizational Ergonomics (Macro-ergonomics)

- Optimisation of **socio-technical systems** – including their organisational structures, policies and processes. Topics: communication, crew resource management, work design, design of working times, teamwork, participatory design, community ergonomics, cooperative work, new work paradigms, virtual organisations, telework, quality management.



Joint optimisation:
neither the technical nor the social part is optimised alone – both together (Tavistock studies – Trist & Bamforth)

Fig 4.6 Socio-technical system

Topic	Key points
Communication	Formal / informal, vertical & horizontal; clarity, feedback, handover (shift change), briefing, signs; avoids errors and accidents
Work design	Job enlargement (more tasks), job enrichment (more responsibility), job rotation (reduce monotony), autonomous work groups, ergonomic task allocation (human vs machine)
Design of working times	Hours of work, rest pauses, shift systems (forward rotation better), night work & circadian rhythm (body temperature lowest ≈ 3–4 am), flexi-time, compressed work week, overtime limits
Teamwork & cooperative work	Clear roles, shared goals, trust; groupware (CSCW), team training, CRM (aviation); social loafing risk
New work programs	Telework / work-from-home, hot-desking, virtual teams, lean production, TQM, participatory ergonomics, safety culture, gig / platform work
Policies & processes	Health & safety policy, incentive schemes, training, staffing levels, management style, participation of workers

Note: Participatory ergonomics: involve workers in identifying and solving ergonomic problems – high acceptance of changes.

8. Environmental Ergonomics

- Concerned with human interaction with the physical environment characterised by **climate (temperature, humidity, air movement), pressure, vibration, noise and light** – and how they affect comfort, health, safety and performance.

(a) Climate / thermal environment

- Heat balance: $M \pm R \pm C - E = S$ (metabolic heat, radiation, convection, evaporation, storage). Four factors: air temperature, humidity, air velocity, mean radiant temperature (+ clothing & activity).
- Thermal comfort zone (sedentary): air temp ≈ 20–26 °C, RH 40–60 %, air speed < 0.2 m/s. Indices: **Effective Temperature (ET), WBGT,** PMV – PPD (Fanger; ISO 7730).
- **WBGT** = 0.7 Tnwb + 0.3 Tg (indoors / no sun); = 0.7 Tnwb + 0.2 Tg + 0.1 Ta (outdoors with sun). Higher WBGT → shorter work period / more rest.
- Heat stress → heat cramps, exhaustion, **heat stroke;** cold stress → hypothermia, frostbite, reduced dexterity (hands < 15 °C). Control – engineering (ventilation, insulation, shields), administrative (work–rest, acclimatisation), PPE.

(b) Pressure

- High altitude (hypobaric): low O2 partial pressure → hypoxia, reduced performance; need acclimatisation / oxygen. Hyperbaric (diving, tunnelling, caissons): decompression sickness ("bends"), nitrogen narcosis – staged decompression is necessary.

(c) Vibration

- **Whole-body vibration** (vehicles, cranes; 0.5–80 Hz): spine resonates at ≈ 4–8 Hz, discomfort, back pain, motion sickness (0.1–0.5 Hz). Standard: ISO 2631.
- **Hand-arm vibration** (power tools; 8–1000 Hz): vibration white finger (Raynaud's), carpal tunnel; ISO 5349 – limit exposure A(8) 5 m/s^2.
- Control: isolation (springs, dampers), anti-vibration handles, seat suspension, reduce exposure time, maintenance, gloves.

(d) Noise

- Sound level measured in dB(A). Permissible exposure ≈ 85 dB(A) for 8 h (NIOSH, with 3-dB exchange rate: halve time for each +3 dB); 90 dB(A) (OSHA / Indian factories norm, 5-dB exchange). Effects: hearing loss (NIHL), annoyance, communication interference, stress, errors.
- Control: at source (quiet machines, damping), path (enclosures, barriers, absorbers), receiver (ear plugs / muffs, job rotation); hearing conservation programme (audiometry).

(e) Illumination / light

- Factors: illuminance (lux), luminance, contrast, glare (direct & reflected), colour, uniformity, age of worker. Poor light → eye strain, headache, errors, accidents.

Task / area	Typical illuminance (lux)
Corridors, stores, passages	50 – 100
Rough work, general factory, machine shops	200 – 300
Office work, reading, computer area	300 – 500
Drawing office, medium assembly, inspection	500 – 750
Fine assembly, instrument work, colour matching	750 – 1000
Very fine work (watch making, micro-electronics)	1500 – 2000 + local light

- Design: use task lighting + general lighting; avoid glare (shielding, matt surfaces, correct angle), ratio of brightness between task : surroundings ≤ 3 : 1 (near), ≤ 10 : 1 (far); colour of light (Ra > 80); maintain lamps.
- Other environmental factors: air quality / ventilation, dust & fumes, humidity, radiation, colour & aesthetics, room layout, hygiene, ergonomic chair / workstation.

9. Application – Ergonomic Workstation Design (Quick Guide)

Item	Guideline
Seated work-surface height	About elbow height when seated (≈ 70–75 cm for keyboard work); adjustable table preferred
Chair	Seat height adjustable (≈ 38–50 cm, popliteal height); seat depth 38–43 cm; lumbar support; backrest 95–110°; castors, arm rests; feet flat on floor / footrest
Monitor / VDT	Top of screen at or slightly below eye level; viewing distance 50–70 cm; line of sight 15–20° below horizontal; no glare; text size ≥ 20 minutes of arc
Keyboard & mouse	Elbows ≈ 90°; wrists neutral (not bent); keyboard at elbow height; mouse close; palm support
Standing work	Precision work: surface 5–10 cm above elbow; light assembly: 5–10 cm below elbow; heavy work: 15–40 cm below elbow height
Breaks	Micro-breaks every 30–45 min; 20-20-20 rule for eyes (every 20 min look 20 feet / 6 m away for 20 s); alternate sit–stand

- General rules: **neutral posture, elbows at sides,** keep frequently-used items within the normal reach area (35–40 cm), heavy items between knee and shoulder height, avoid static positions, vary tasks, provide adjustability and good lighting.
- Ergonomic evaluation process: identify task → collect data (observation, video, checklists, questionnaires, body-part discomfort map) → analyse risk (RULA / REBA / NIOSH) → redesign (engineering controls first, then administrative, PPE last) → implement → evaluate and follow-up.

Risk factor	Typical ergonomic control
Excessive force / lifting	Mechanical aids, lighter loads, team lifting, better handles & grip, adjustable height tables
Repetition	Automation, job rotation, rest pauses, power tools, task enlargement
Awkward / static posture	Adjustable workstation, re-position work, supports, tilting tables, sit-stand
Vibration / cold / contact stress	Anti-vibration tools, gloves, padding, rounded edges, warm environment

10. Quick Revision & Probable Questions

Memory hooks

- Ergonomics = ergon + nomos; fit the job to the person. Domains: Physical – Cognitive – Organizational – Environmental.
- Disciplines: Psychology, Engineering, Biomechanics, Industrial design, Graphic design, Statistics, OR, Anthropometry.
- Percentiles: 5th = mean – 1.645σ; 95th = mean + 1.645σ. Extremes: clearance → 95th; reach → 5th.
- NIOSH: $RWL = 23 \times HM \times VM \times DM \times AM \times FM \times CM$; LI = load / RWL.
- Murrell: $R = T\,(W - S)/(W - 1.5)$. $WBGT = 0.7\,Tnwb + 0.3\,Tg$.

1. Define ergonomics. Explain its history, objectives and the disciplines contributing to it.
2. Explain design morphology and its relationship with the cognitive abilities of human beings.
3. What is anthropometry? Explain the principles of applying anthropometric data with examples.

4. Explain physical ergonomics – anatomy, physiology and biomechanics related to physical activity.
5. Discuss cognitive ergonomics: perception, memory, reasoning, mental workload and decision-making.
6. What is organisational ergonomics? Discuss work design, working times and teamwork.
7. Explain environmental ergonomics – climate, noise, vibration, illumination and pressure.

UNIT – V

Man–Machine Interaction • Displays • Controls • Design Guidelines • Standards

Syllabus (as given)

Man–Machine Interaction: man–machine interaction cycle, man–machine interfaces, displays – factors that control choice of display, visual displays – qualitative displays: moving pointer displays, moving scale displays, digital displays, indicators, auditory displays, tactile displays. Factors affecting effectiveness of displays. Quantitative displays, check-reading displays, representational displays. Types of controls and their integration with displays. Design guidelines for displays and controls: viewing distance, illumination, angle of view, reach etc.; general design checklist for displays and controls. Standards for ergonomics in engineering and design, displays and controls.

1. Man–Machine System and the Interaction Cycle

- A **man–machine system** (human–machine system) is a combination of one or more humans and one or more machines (equipment, tools, computers, vehicles) which interact to perform a function in a given environment. Examples: driver–car, operator–lathe, pilot–aircraft, user–smartphone.
- Types by degree of human control: **Manual** (human supplies power & control – hand tool), **Mechanical / semi-automatic** (machine gives power, human controls – lathe, car), **Automatic** (machine does sensing, decision & action; human monitors & maintains – CNC, robot, process plant).
- Why study? Most accidents & errors are due to poor interface design; system performance = performance of the human + machine + the way they are linked.

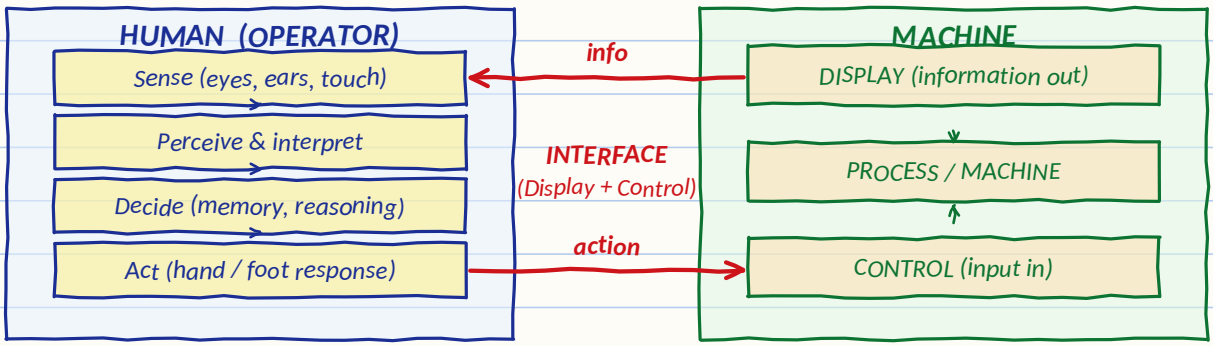


Fig 5.1 Man–machine interaction cycle

- **Interaction cycle:** Machine state shown on **display** → operator **senses** the signal → **perceives** & interprets using memory → **decides** what to do → **acts** on the **control** → machine changes state (process) → new output on display (feedback) → the cycle repeats. It is a **closed-loop** system; time delay / lag in any part reduces performance.
- **Man–machine interface:** the point(s) where information / energy is exchanged – **displays** (machine → man), **controls** (man → machine), plus the work-place layout, environment, software / GUI, manuals, communications.

Humans are better at	Machines are better at
Perceiving patterns, judging, improvising, flexibility	Fast, repetitive, precise work; high force & power
Handling unexpected events, inductive reasoning	Storing & recalling huge data accurately; deductive logic
Sensitive detection of small signals; selecting	Monitoring for long periods without fatigue (vigilance)
Creativity, ethics, value decisions	Operating in hazardous environment; simultaneous tasks

Note: Function allocation ("MABA-MABA", Fitts list): assign each function to the one – human or machine – that does it better; keep the human in control (supervisory role).

2. Displays – Functions, Classification and Choice

- **Display** = a device that presents information about the machine / process to the operator (by sight, sound or touch). Functions: give **status** information, **quantity,** warn of danger, instruct, show trend, help in control (tracking), and identify.

Classification basis	Types
Sense used	**Visual, auditory, tactile** (haptic) displays
Nature of information	**Quantitative** (numerical value), **qualitative** (status / trend / direction), **check-reading** (normal or not), **representational** (pictorial / symbolic), alphanumeric, warning / indicator
Time variation	Static (signs, charts, labels) and dynamic (dial, CRT, LED, LCD)
Technology	Mechanical (analogue dial), electronic (digital, LED, LCD), screens, projection, HUD, VR / AR

Factors that control the choice of display

1. **Type of information** to be shown (exact value → digital / quantitative; trend / rate → analogue; status → light; warning → sound).
2. **Accuracy and speed** of reading required; **resolution:** response time.
3. Task of the operator: reading, tracking, check reading, setting, monitoring; need for rate-of-change and trend.
4. **Environment:** illumination level, noise, vibration, dust, temperature, glare, darkness.
5. **Space available** on the panel; distance & angle of viewing; number of displays; operator position and mobility.
6. Operator characteristics: visual acuity, training, age, expectations (stereotypes); other loads on the sense (visual overload → use auditory).
7. Cost, reliability, maintainability; compatibility with controls and with other displays.

Use VISUAL display when	Use AUDITORY display when	Use TACTILE display when
Message is complex, long, or needs later reference; relates to location in space; auditory channel overloaded; environment noisy; person stays at one place	Message is simple, short & urgent (warning); calls for immediate action; person moves about; visual channel overloaded; dark / poor lighting; no need for later reference	Both visual & auditory channels are busy or unusable (dark, noise); for identification of controls by feel; blind users (Braille); alerts (vibration)

3. Visual Displays

(a) Qualitative displays

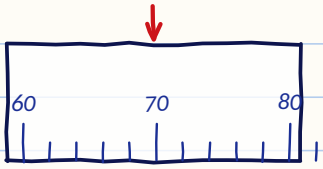
- Give **approximate / non-numerical information:** status (on–off, safe–danger), condition (hot–normal–cold), trend, rate of change or direction of change, rough value. Use coloured zones or simple symbols on the dial (e.g. green = normal, yellow = caution, red = danger).
- Examples: a fuel gauge (E – ½ – F), engine-temperature gauge with a green band, tyre-pressure indicator with limits, signal lamp.
- Advantage: fast reading, a glance shows whether things are normal; less chance of error; takes less space.

(b) Quantitative displays – dynamic types

- Give **exact numerical value** of a variable (speed, temperature, pressure, time). Three main types:



(1) Moving pointer, fixed scale



(2) Moving scale, fixed pointer (window)
(scale slides behind pointer)



(3) Digital display (counter / LED / LCD)

Fig 5.2 Types of quantitative visual displays

Characteristic	Moving pointer – fixed scale	Moving scale – fixed pointer	Digital (counter)
Quantitative reading	Fair (needs scale reading)	Good if scale not too long; poor if moving fast	Best – fastest & most accurate for exact value (fewest errors)
Qualitative / trend / rate-of-change	Good – movement of pointer gives rate & direction	Poor – the moving scale is hard to follow	Poor – numbers jump; no rate cue
Setting (matching a value)	Good – control–pointer relation clear	Fair	Good if counter turned directly by knob
Tracking	Good – best	Poor	Poor
Space needed	Large (long scale)	Small (only window shown) – long range in little space	Very small
Check reading	Good – pointer position gives quick cue	Poor	Poor

Design of scales & pointers (quantitative visual displays)

- **Scale length** matched to the required accuracy (more divisions for more precise reading); open (partial) circular or horizontal scales are preferred; vertical scales read in the same direction as the change (up = increase).
- Graduations: **major marks numbered, with minor marks between;** number progression by 1, 2, 5, 10 or multiples (avoid 3, 4, 6, 7, 8 as scale increments); minimum 1 : 2 or 1 : 5 sub-division; do not use more than 9–10 minor marks between numbers; avoid fractions.

- Numerals increase **clockwise** (circular), left → right (horizontal), bottom → top (vertical); numbers upright and **outside** the scale marks (not hidden by pointer); numerals in simple type face (sans-serif), black on white (or white on black) for contrast.
- **Pointer:** simple, tapered, tip touches the smallest graduation (no gap, no overlap); close to the dial face (to avoid parallax); same colour as marks; counter-balanced; length reaches the smallest mark.
- Colour coding of dial zones (green / yellow / red) for qualitative information; marks at the normal operating value.
- **Parallax** error: due to gap between pointer and scale – viewer's eye not perpendicular; reduce with small gap, mirrors on the scale.

(c) Check-reading displays

- Used to find quickly whether a value is **normal or deviating** – e.g. in panels of many identical dials: all pointers are arranged in a line (e.g. at the "9 o'clock" or "12 o'clock" position); any deviation is easily seen by the change in pattern. Quantitative reading not needed.

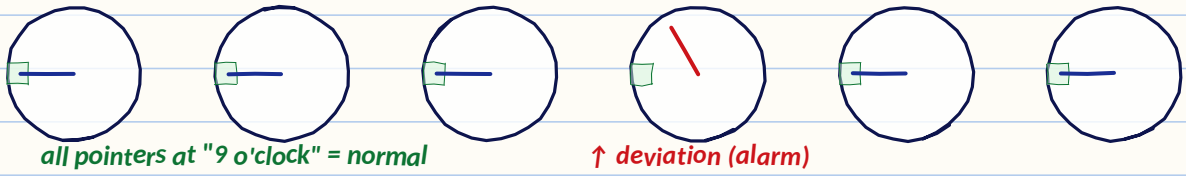


Fig 5.3 Check-reading (comparison) display panel

(d) Representational (pictorial) displays

- Show the **relationship / situation as a picture or symbol** so that the operator understands the system "at a glance": aircraft attitude (artificial horizon) indicator, map / navigation display (GPS), **mimic diagrams** of a power plant or pipe network, HUD in cars / aircraft, CAD or SCADA graphical screens, radar displays.
- Principles: **pictorial realism** (looks like the real thing), **moving-part principle** (moving element moves like the real thing), **principle of the colocated** / proximity compatibility (related information close together), minimise clutter, de-clutter by layers.

(e) Indicators / signal lights (qualitative status displays)

- Used for **status / warning / alarm** (on-off, ready, fault). Design: size (visual angle ≥ 1°), brightness greater than surroundings (not dazzling), position within central visual field, flashing for urgency (3–10 flashes / s; "on" time ≥ 0.05 s; flash rate 1–2 Hz for attention; flashing lights are intrusive – use only for urgent condition), and legends beside the lamp.

Colour	Meaning (IEC 60073 / common practice)	Typical use
Red	Danger, emergency, stop, fault, abnormal – needs immediate action	Emergency stop button, over-temperature, alarm
Yellow / Amber	Caution, attention, abnormal condition developing, near limits	Pre-warning, low level
Green	Normal, safe, ready, "go"	Machine running, supply on
Blue	Mandatory action; advisory information	Reset / required operator action
White / grey	Neutral, general information; no specific meaning	Power on, auxiliary

Note: Do not use more than ≈ 5 colours; avoid colour as the only cue (colour-blind ≈ 8 % of men) – back up with position, shape or label.

4. Auditory Displays

- Convey information by **sound:** warning, alarm, signal, speech message, feedback of key press. The ear can receive from all directions, is always "open", and responds faster (≈ 0.15 s) than the eye – good for urgent warnings.

Type	Characteristics / use
Pure tones / beeps	Single frequency; feedback, "ready", simple alert; may be masked by noise
Complex tones, chimes, gongs	Several frequencies; pleasant, e.g. station announcements, appliance alerts
Buzzers & bells	Low-frequency harsh buzz – equipment failure; bells – telephone, fire, school; high intensity
Horns, sirens	Loud, rising–falling pitch – emergency / evacuation, vehicles; detectable at a distance
Speech (voice) messages	Give complex information (aircraft "pull up"; train announcement); needs good intelligibility & language
Auditory icons / earcons	Short, symbolic sounds representing events (computer, mobiles, cars)

- Design guidelines:
  - Frequency in the range **200 – 5000 Hz,** best 500 – 3000 Hz (ear most sensitive); for long distance and through obstacles use < 1000 Hz.
  - Intensity at least **10 – 15 dB above the masked threshold** of the background noise at the operator's ear (but < 120 dB, avoid startle / hearing damage); allow volume control for non-critical sounds.

- Make the signal **distinct** from the background noise & from other signals (different frequency, interrupted / modulated; 0.2 – 0.5 s beeps, 1–8 / s); use **intermittent** rather than continuous tones; ensure the sound is meaningful & learnt; use maximum 4–5 different warning tones.
- Ensure signal is audible everywhere in the area (avoid reflection); do not use if the ambient noise is extremely high or the listener is wearing hearing protection; provide a silence / acknowledge switch.
- Use a combined visual + auditory warning for critical alarms (redundancy), and give priority levels (urgent = faster pulses, higher pitch).

5. Tactile (Haptic) Displays

- Information given through the **sense of touch** – skin receptors (pressure, vibration, temperature). The fingertips are most sensitive (two-point discrimination ≈ 2 mm).
- Uses: **shape / size / texture coding of controls** (so a pilot or driver can identify them without looking – flap lever has flap-shaped knob, landing-gear lever has wheel-shaped knob); **Braille** and raised graphics for blind users; vibration alerts (mobile phone, steering wheel, joystick – "haptic feedback"); pin-matrix devices; force-feedback in tele-operation / surgery robots & virtual reality.
- Advantages: not affected by darkness, noise or glare; private; leaves eyes / ears free. Limitations: low information rate; needs skin contact; affected by gloves, cold, vibration; limited coding (≈ 6–10 shapes).
- Guidelines: textures should be clearly different (smooth, knurled, fluted); shapes distinguishable with gloves; vibration frequency ≈ 150–300 Hz (most sensitive: ≈ 250 Hz); use **at least 3 mm height difference** / distinct size steps for size coding; consistent meaning.

6. Factors Affecting Effectiveness of Displays

Factor	Effect / design remedy
Contrast & brightness	Low contrast reduces legibility; keep contrast ratio ≥ 3 : 1, preferably 5 – 10 : 1 between figure and ground; black on white / white on black
Illumination & glare	Too low → eye strain; too high / glare → reflections on glass cover; use matt finish, shield, correct level 200 – 500 lux
Size of characters / viewing distance	Visual angle ≥ 20 – 22 minutes of arc; increase for poor illumination, vibration, fast reading

Factor	Effect / design remedy
Angle of view & position	Within the normal visual field (30° cone); avoid parallax; normal to the line of sight
Colour & coding	Use standard / stereotype colours; avoid too many; maintain colour contrast; not only colour
Clutter / information load	Too many displays or signals hamper detection; group & prioritise; remove unnecessary data
Operator factors	Visual acuity, age, colour vision, training, expectation, fatigue, stress
Environment	Vibration, glare, ambient noise (for auditory), dust, temperature; operating speed / time of exposure
Compatibility	Display and control relation, population stereotypes, language, symbols, consistent layout

7. Controls – Types, Coding and Integration with Displays

- **Control** = a device through which the operator transmits an **input / command** to the machine (hand, foot, voice, eye). Functions: **activate / turn on-off, discrete setting** (select among positions), **continuous setting / adjust, data entry, tracking / steering.**

Type of control	Examples	Used for
Discrete – two position	Push button, toggle switch, rocker, key switch, foot switch	On / off, start / stop, emergency stop
Discrete – multi-position	Rotary selector, thumb-wheel, slide / detent switch, push-button keypad	Selection of mode / speed / channel (3 – 24 positions)
Continuous – rotary	Knob, hand-wheel, crank, steering wheel	Fine setting (volume), steering, large force (hand-wheel)
Continuous – linear	Lever (hand / joystick), slider, push-pull knob, pedal	Gross positioning, speed (throttle), brake, tracking
Data-entry	Keyboard, touch screen, mouse, voice input	Text, command, pointing

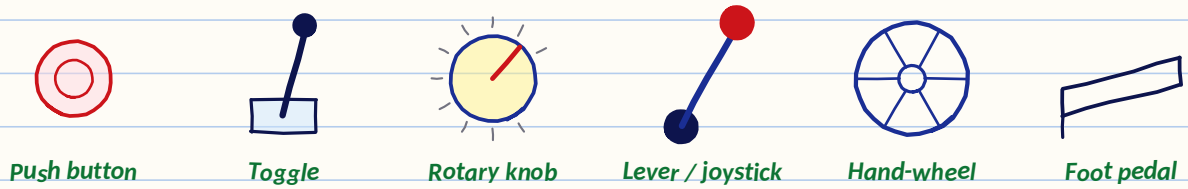


Fig 5.4 Common types of controls (sketch)

Coding of controls (so that they are not confused)

- **Shape** (best – can be identified by touch), **size** (3 – 4 sizes), **colour** (red for emergency / stop), **location / position** (fixed place, standard layout), **texture / surface** (knurled, smooth), **mode of operation** (push, pull, turn), **labelling** (word or symbol – needs vision & light). Combine two or more codes for critical controls.
- **Resistance of controls:** elastic (spring) – gives feedback, good for fine setting; static friction (stiction); viscous damping (smooth, stable); inertia (flywheel, for steady motion). Resistance prevents accidental activation and gives "feel".
- Prevent accidental operation of critical controls: recessed or **guarded / covered,** interlock / lock, sequence of two actions, enough resistance, separation, orientation; emergency stop – large red mushroom head, easy to reach, push-to-lock / twist-to-release.

Control–display relationships (integration)

- **Compatibility:** movement of control and response of display / machine should match user expectation – **spatial** (location), **movement** (direction), **conceptual** (symbol meaning), modality. Stereotypes: clockwise / up / right / forward = increase, on, start; counter-clockwise / down / left / back = decrease, off.
- **Warrick's principle:** the pointer of the display should move in the **same direction as the nearest point on the control** (for a knob next to a dial: clockwise turn moves the nearest part of the knob and the pointer in the same direction).
- **Control–display (C/D) ratio** = (movement of control) / (movement of display) (e.g. degrees of knob turn per cm of pointer travel). Low C/D → quick but coarse; high C/D → slow but fine. There is an **optimum C/D ratio** that minimises the total time = gross movement time + fine adjustment time (for knobs ≈ 0.2 – 0.8 depending on size / task).
- **Gain** = display change / control change (reverse of C/D); lag (time delay) and backlash should be minimised.

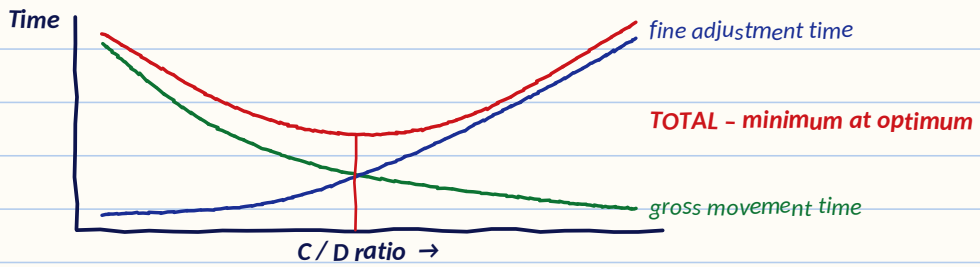


Fig 5.5 Optimum control–display ratio

Integration / arrangement of displays and controls on a panel

- Arrange by **importance and frequency of use** (most important & frequent in the best position – central visual field and easy reach); by **function** (related items together, outlined / colour-grouped); by **sequence of use** (left → right, top → bottom); by **control–display link** (each control placed close to / below its display; use same layout, labels, colours; avoid hand covering the display – place control below or to the right of the display for right-handed users).
- Natural mapping: layout of controls mirrors layout of the controlled things (e.g. four stove burners vs four knobs – put knobs in same 2×2 pattern); consistency and standardisation (e.g. the same layout in all vehicles).

8. Design Guidelines for Displays and Controls

(a) Viewing distance

- Normal viewing distance for dials / panels ≈ **50 – 70 cm** (not less than ≈ 30 cm – accommodation limit; VDT 50 – 70 cm). The size of characters must increase in proportion to distance.

Character height from visual angle

$H = (D \times \theta) / 3438$ H = height of character, D = viewing distance (same units), θ = visual angle in minutes of arc (1 radian = 3438 minutes).

Recommended visual angle: 20 – 22 min (min. 16 min for good conditions); 30 – 45 min for poor illumination / important / fast-reading tasks.

Example: D = 700 mm, θ = 20 min → H = 700 × 20 / 3438 = **4.07 mm** (≈ 4 mm); for θ = 40 min H ≈ 8.1 mm.

(b) Illumination

- General level 200 – 500 lux for reading dials and panels (higher, ≈ 500 – 1000 lux, for fine detail); uniform; no glare, reflection or shadows on the face of the display – matt glass, hoods, correct position of luminaires.
- For night / dark-room control (aircraft, ships, control rooms): **red or low-level white light** to keep dark adaptation; individually adjustable (dimmer) back-lighting; luminance of self-lit display not > 3 times the surrounding.

(c) Angle of view

- The normal line of sight is about **15° below the horizontal** (in standing) and **≈ 30° – 35° below horizontal** for seated, relaxed head / eye position; displays should lie within **30° cone** around it (critical ones within ≈ 15°); maximum 60° with head movement.

- Horizontal: optimum ± 15° from the median plane; max. eye rotation ± 35°; head turn up to ± 95° (avoid). Mount displays **perpendicular to line of sight** (± 15° tilt) to avoid parallax and reflection.

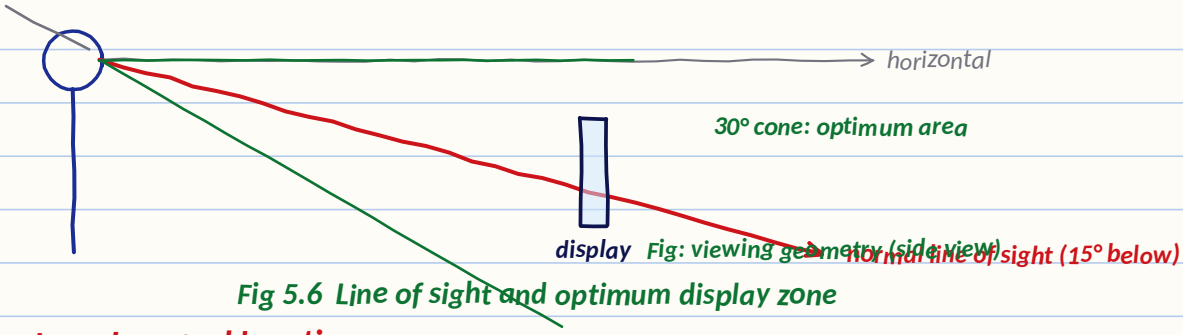


Fig: viewing geometry (side view)

Fig 5.6 Line of sight and optimum display zone

(d) Reach and control location

- Locate frequently used and important controls in the **normal working area** (forearm sweep ≈ 35 – 45 cm) and within the **maximum reach** (arm sweep ≈ 55 – 65 cm for 5th percentile female); occasionally used controls further away; heavy / large forces – hand-wheels / pedals at suitable height (≈ 90 – 110 cm for hand controls, elbow height; foot controls – ankle/knee height ahead).
- Use the **5th percentile reach** to decide the maximum distance and the 95th percentile for clearances; sitting reach differs from standing – consider clothing and seat position; leave space for gloves.
- Typical dimensions (check standards): push-button diameter 10 – 25 mm (finger), spacing ≥ 15 mm (10 mm for keyboards); toggle switch tip 3 – 25 mm, spacing 25 – 50 mm; knob diameter 10 – 100 mm, depth 12 – 25 mm; hand-wheel 175 – 525 mm; pedal 25 – 30 × 8 – 10 cm, travel 5 – 15 cm, force 20 – 90 N; emergency push-button ≥ 25 mm, red mushroom head.

9. General Design Checklist – Displays and Controls

Displays – check	Controls – check
Is the right type of display used for the information (quantitative / qualitative / check reading)?	Is the right control type used for the task (discrete / continuous; hand / foot; force & precision)?
Is it visible and legible from the usual positions (distance, angle, light)? Character height ≥ 20 min arc?	Is it within reach of 5th percentile user, not obstructing the view and displays; correct height?
Are scales simple (1, 2, 5, 10), numerals upright, pointer touches marks, no parallax?	Are controls coded (shape, size, colour, label) & distinguishable; spacing sufficient for gloves?
Is the display grouped by function / sequence / importance; critical items in central 30° zone?	Are control–display relations compatible (stereotypes, Warrick, C/D ratio, same layout)?

Displays – check	Controls – check
Are warnings coded (red / amber / green), redundant (visual + auditory) & not too many?	Is resistance suitable (feedback, no accidental activation, guards for emergency / dangerous)?
Is glare, reflection, vibration, noise, darkness considered? Night lighting available?	Is the movement direction as expected (clockwise = increase)? feedback (click / light) provided?
Is the information load low (no clutter), consistent with standards and user training?	Can errors be reversed (undo / recover)? Is there an emergency stop & reset sequence?
Is maintenance (cleaning, replacement of lamps) easy; failure of display detectable?	Is the control robust, easy to clean, usable with PPE; tested by users?

10. Standards for Ergonomics in Engineering Design

- Standards give **minimum requirements, dimensions, procedures and test methods** so that systems are safe and usable. They are developed by ISO, IEC, CEN (EN), BIS (India), ANSI / HFES, military (MIL-STD) and national bodies; they are revised periodically – always refer to the latest edition.

Standard	Scope (ergonomics)
ISO 6385	Ergonomic principles in the design of work systems (general principles)
ISO 9241 series (e.g. 9241-210, -11)	Ergonomics of human–system interaction: usability, human-centred design, display, keyboard, dialogue
ISO 7250	Basic human body measurements for technological design (anthropometry)
ISO 14738 / EN 547	Anthropometric requirements for the design of workstations at machinery (body dimensions)
ISO 9355 / EN 894	Ergonomic requirements for the design of displays and control actuators (parts 1–3)
ISO 11064	Ergonomic design of control centres (layout, workstations, displays, environment)
ISO 7731 / ISO 11428	Auditory danger signals / visual danger signals (design, detectability)
IEC 60073 / IEC 60447	Colours of indicator lights and push-buttons; actuating principles (coding)
ISO 11228 (parts 1–3)	Manual handling – lifting, carrying, pushing / pulling, repetitive tasks
ISO 7730, ISO 2631, ISO 5349	Thermal comfort (PMV / PPD); whole-body vibration; hand-arm vibration
ISO 15534 / ISO 11226	Dimensions for openings (access); evaluation of static working postures
ANSI / HFES 100, 200; MIL-STD-1472	Human factors engineering of computer workstations & software; human engineering design criteria for military systems

Standard	Scope (ergonomics)
BIS (India): IS 3646 etc.	Indian standards, e.g. interior illumination (IS 3646) & others; Factories Act norms for noise / lighting / ventilation
OSHA / NIOSH guidelines	Workplace safety; lifting equation; noise & heat limits; ergonomic programmes

Note: In answers, quote the principle of the standard (e.g. "ISO 9355 – coding & arrangement of displays & controls") rather than clause numbers.

11. Quick Revision & Probable Questions

Memory hooks

- Cycle: Display → Sense → Perceive → Decide → Act → Control → Machine → (feedback).
- Visual displays: moving pointer / moving scale / digital; qualitative – quantitative – check-reading – representational.
- Visual: complex, permanent, quiet; Auditory: simple, urgent, moving person, dark; Tactile: eyes & ears busy.
- $H = D \times \theta / 3438$; θ = 20 – 22 minutes of arc; line of sight 15° below horizontal; C/D ratio optimum.
- Red = danger; Amber = caution; Green = normal; Blue = mandatory; White = neutral.

1. Explain the man–machine interaction cycle with a neat sketch. What are man–machine interfaces?
2. Discuss the factors that control the choice of a display. Compare visual, auditory and tactile displays.
3. Explain moving pointer, moving scale and digital displays with merits & demerits. How is a scale designed?
4. What are check-reading and representational displays? Explain indicators and colour coding of signal lights.
5. Explain the factors affecting effectiveness of displays.
6. Classify controls; explain coding of controls and their integration with displays (compatibility, C/D ratio).
7. Explain design guidelines for displays & controls: viewing distance, illumination, angle of view, reach. Give a general checklist.

8. Name the important standards for ergonomics in engineering design and displays & controls.
9. Numerical: find character height for $D = 700$ mm and $\theta = 20$ min.